

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

**Building 7
Ground floor
Room 736**



**Prof. Wafaa
Professor of Anatomy &
Embryology**



Brachial Plexus Anatomy and Clinical Correlations

Lectures 14 and 15

Prof. Wafaa Saadeh

Learning Objectives

Anatomical Structure

Understand the formation and organization of the brachial plexus from roots to terminal branches

Nerve Pathways

Identify the course and distribution of major nerves arising from the plexus

Clinical Applications

Recognize lesion patterns and their effects on motor and sensory function

Diagnostic Skills

Correlate anatomical knowledge with clinical presentations and deformities

Brachial Plexus

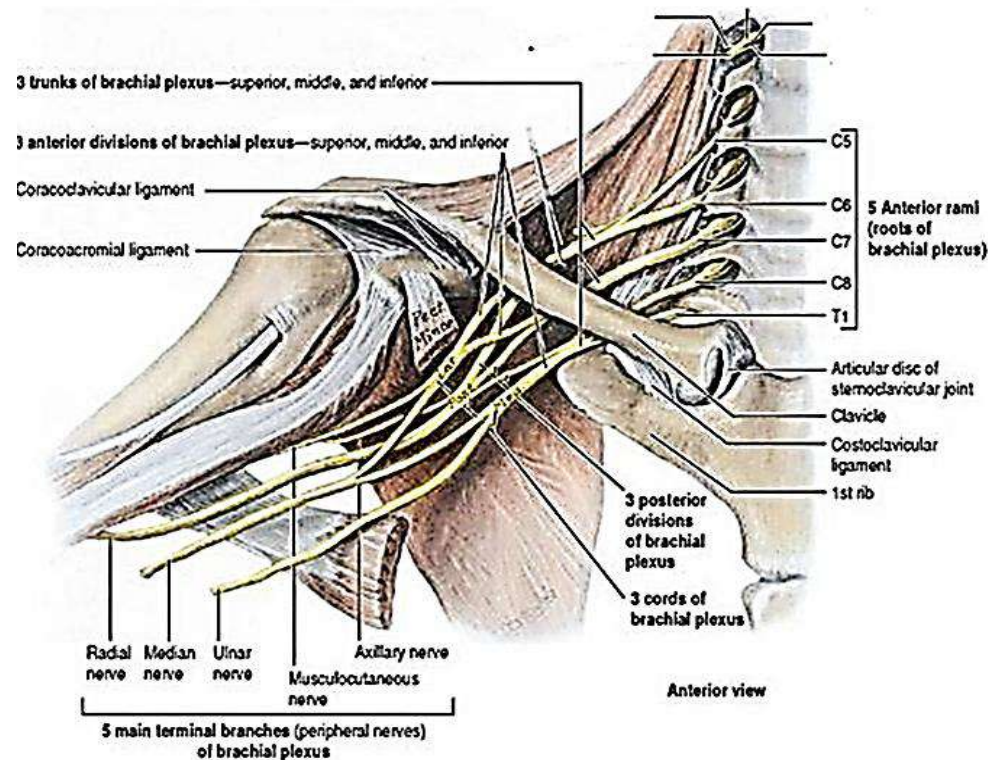
Definition: It is a network of nerves responsible for nerve supply of UL.

Formation:

Anterior rami of C 5, 6,7,8 and T1

prefixed plexus: It may take origin from C4

postfixed plexus: It may take origin from T2



Organizational structure: from roots to branches

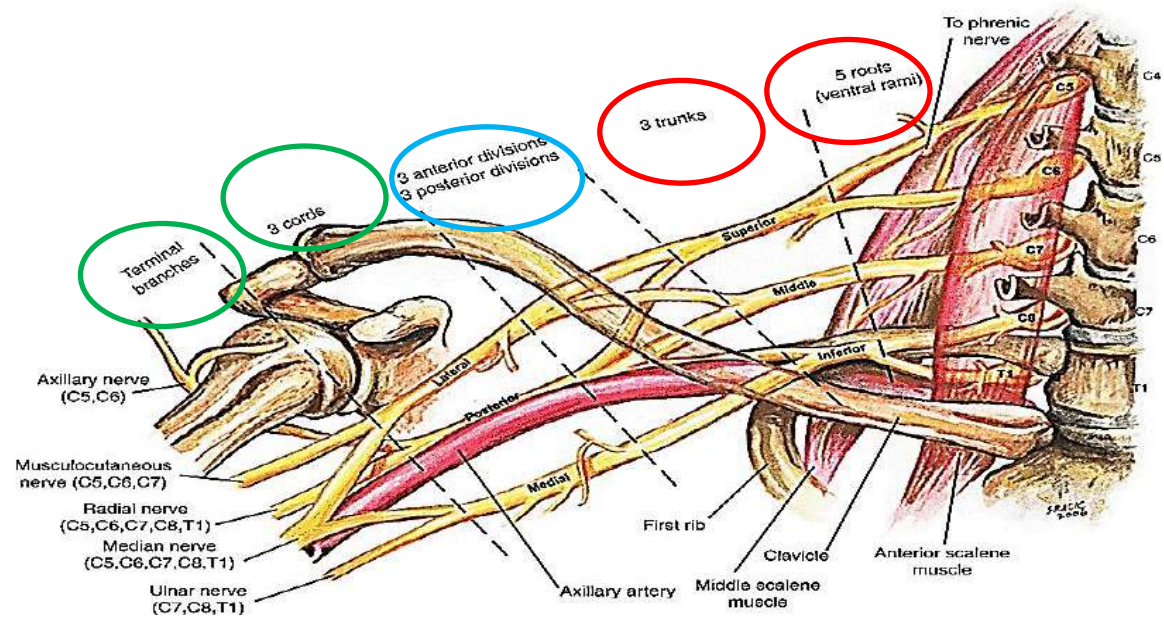
Roots

Trunks

Divisions

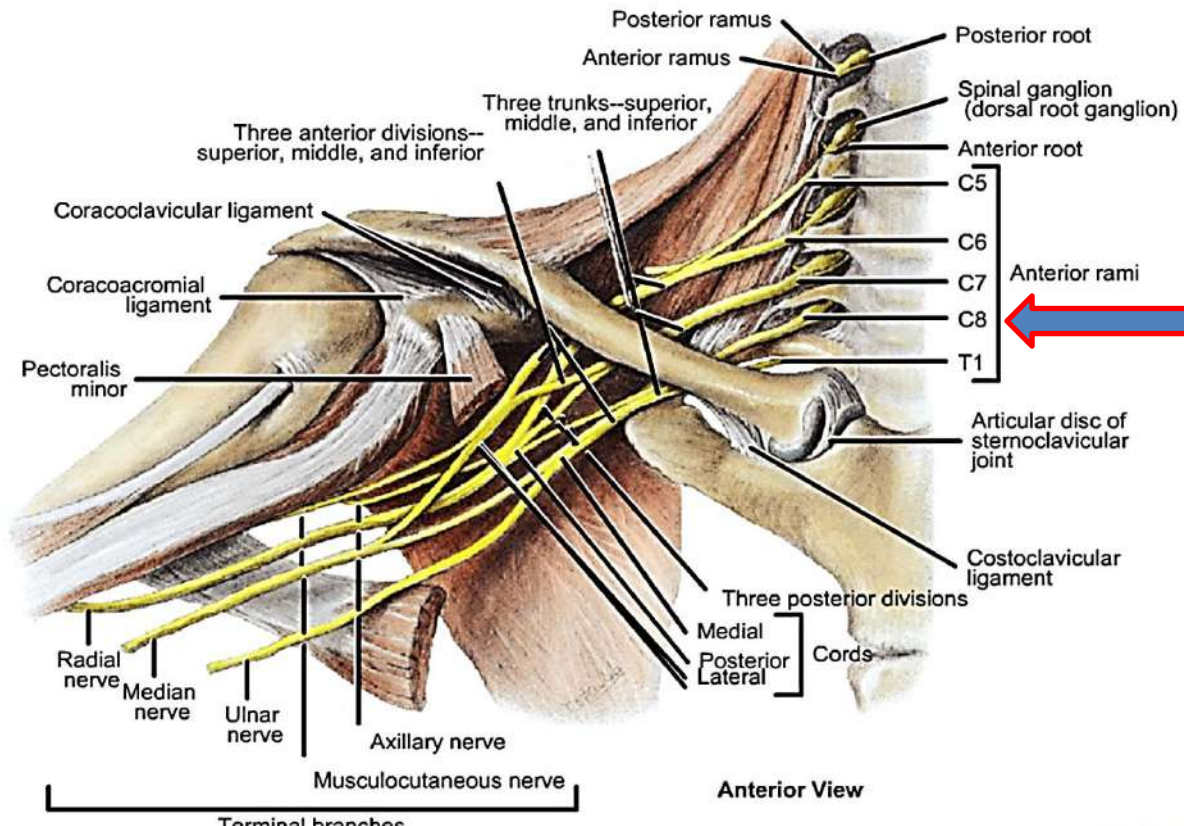
Cords

Branches



Roots

- They are formed by the ventral rami of the cervical 5th, 6th, 7th, 8th and thoracic T1 spinal nerves.
- They lie in the posterior triangle of the neck.
- They unite together to form the trunks

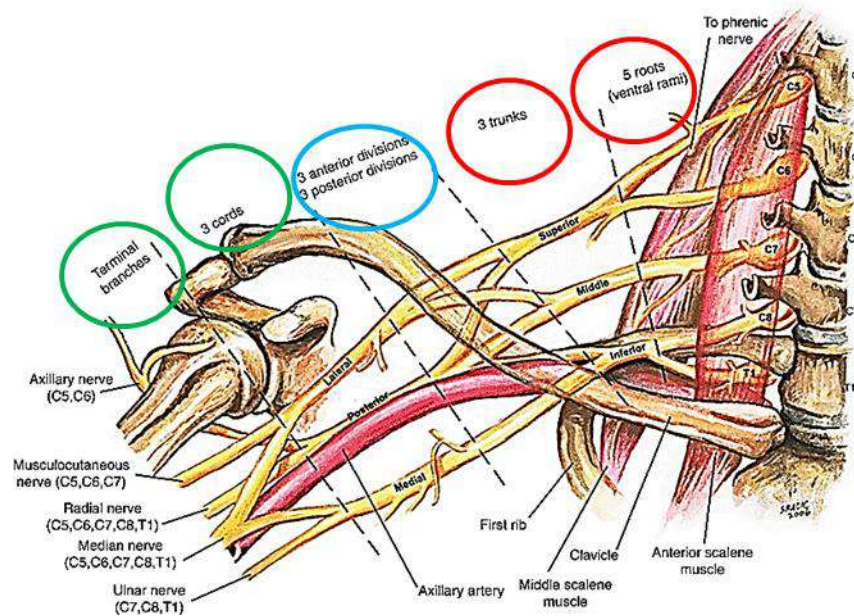


Trunks

- There are 3 trunks: upper, middle and lower.
- They are formed by the roots of the plexus.

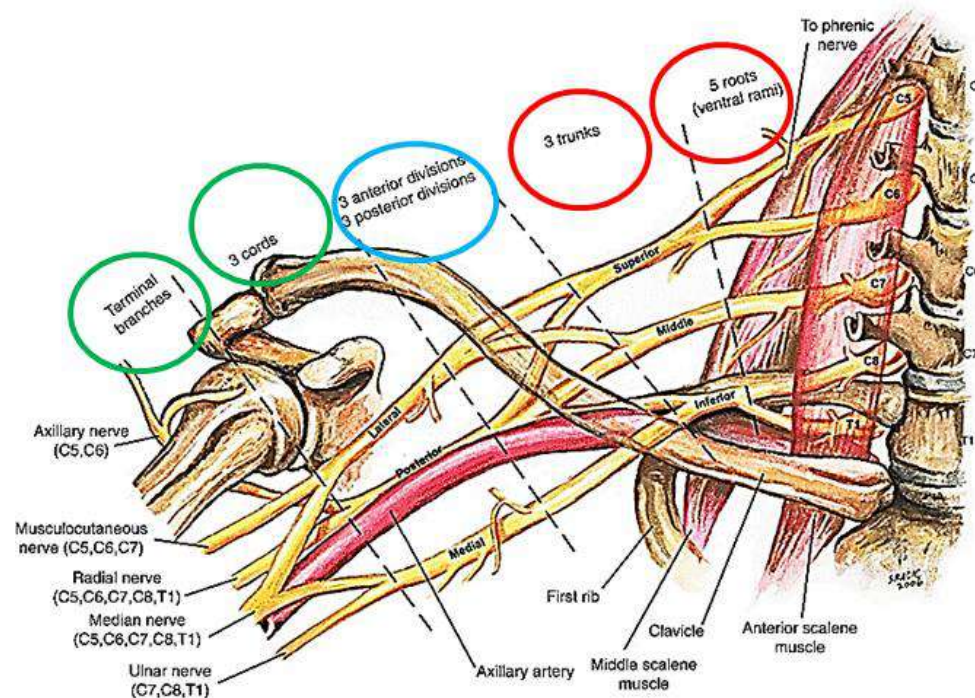
Formation:

- **Upper trunk** : ventral rami of the 5th + 6th cervical nerves.
 - **Middle trunk**: ventral ramus of the 7th cervical nerve only.
 - **Lower trunk**: ventral rami of the of 8th cervical + 1st thoracic nerves.
- They lie in the posterior triangle of the neck.
 - Each trunk divides into : anterior and posterior division.



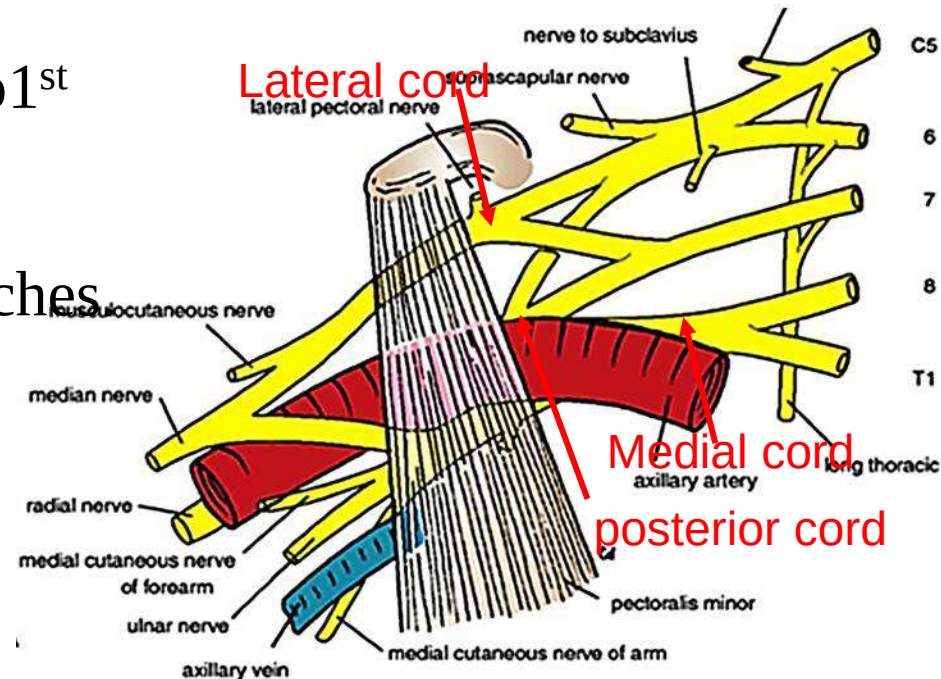
Divisions

- The division occur in the trunks.
- The trunks form 6 divisions (3 anterior and 3 posterior).
- They lie behind the middle third of the clavicle (at apex of the axilla).
- The divisions unite to form the cords of the brachial plexus.



Cords

- There are 3 cords: lateral, medial and posterior.
- **Lateral Cord:** is formed of the anterior divisions of the **upper + middle trunks (C5,6,7)**.
- **Medial Cord** is formed by the anterior division of the **lower trunk alone (C8, T1)**.
- **Posterior Cord** is formed by the posterior divisions of the **upper + middle + lower trunks (C5,6,7,8, T1)**.
- They lie in the axilla (in relation to 1st and 2nd parts of axillary artery).
- Each cord gives off terminal branches.



Relations

Relations of the cords to the axillary artery in the axilla

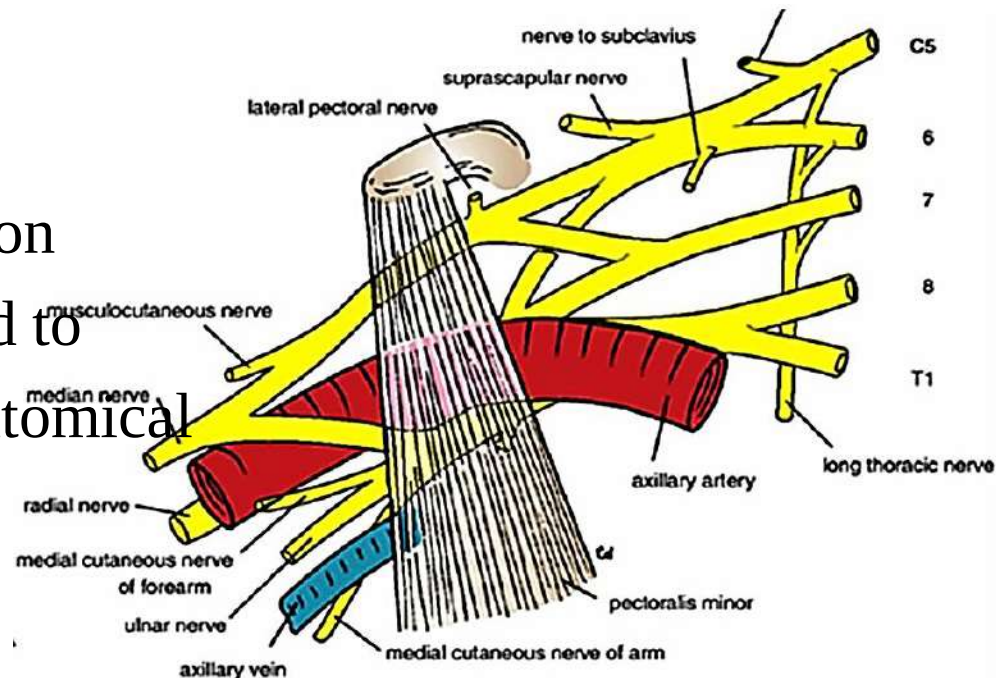
- The **lateral & posterior cords** lie above and lateral to **1st part** of axillary artery.
- The **medial cord** lies posterior to the **1st part** of axillary artery
- **Cords** lie in their anatomical position in relation to **2nd part** of axillary artery.

Lateral cord is lateral in position

Medial cord is medial in position

Posterior cord is posterior in position

- **Branches** of the cords are related to the **3rd part** according to their anatomical position.



Branches of the Brachial Plexus

From the

- Roots
- Trunks
- Cords

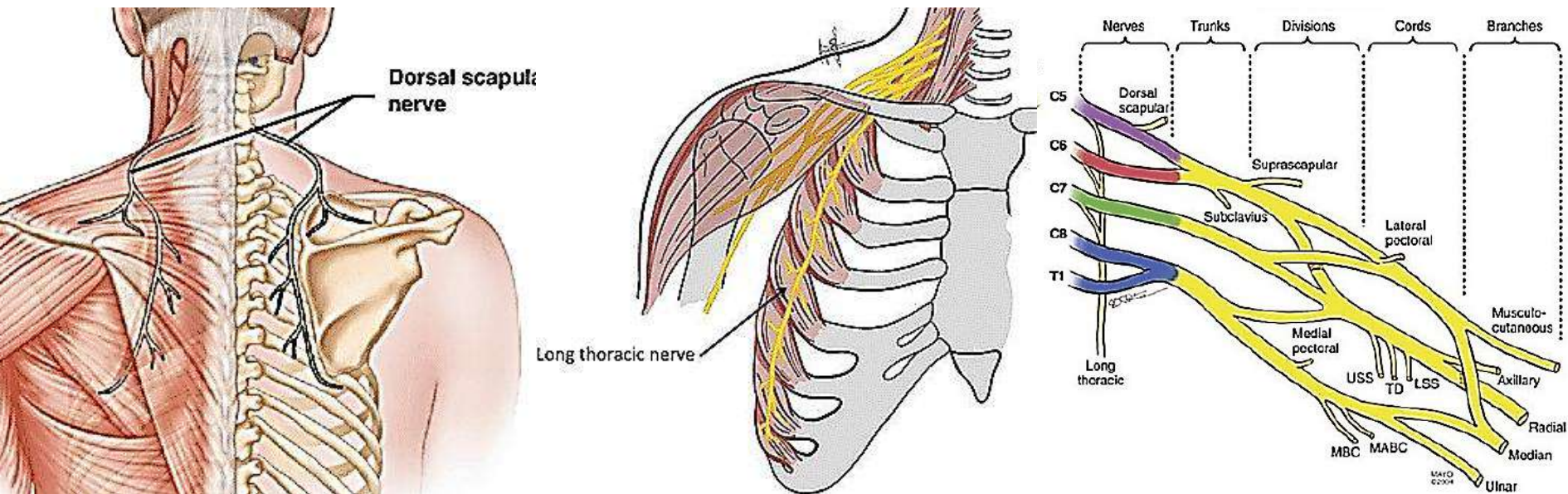
Branches from roots

Dorsal scapular nerve (C5)

supplies the levator scapulae and rhomboid muscles (major and minor).

Long thoracic nerve (C5,6,7)

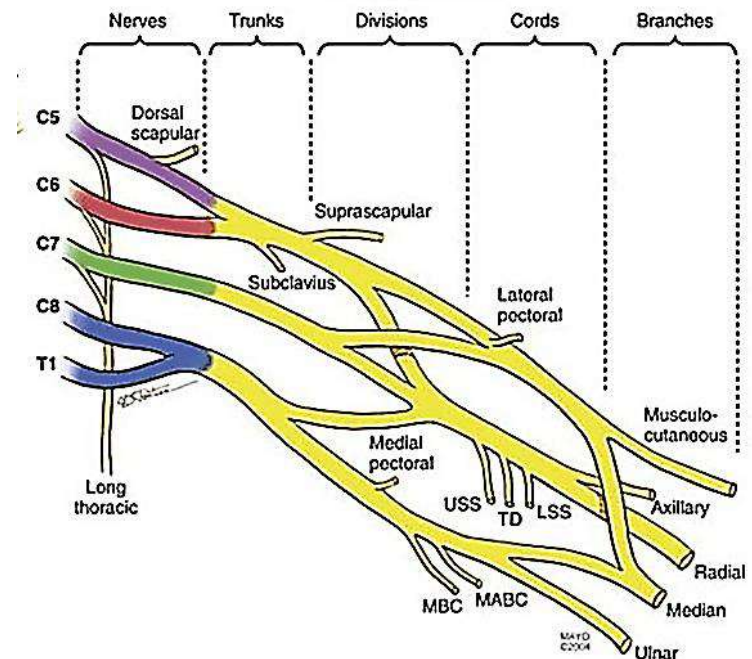
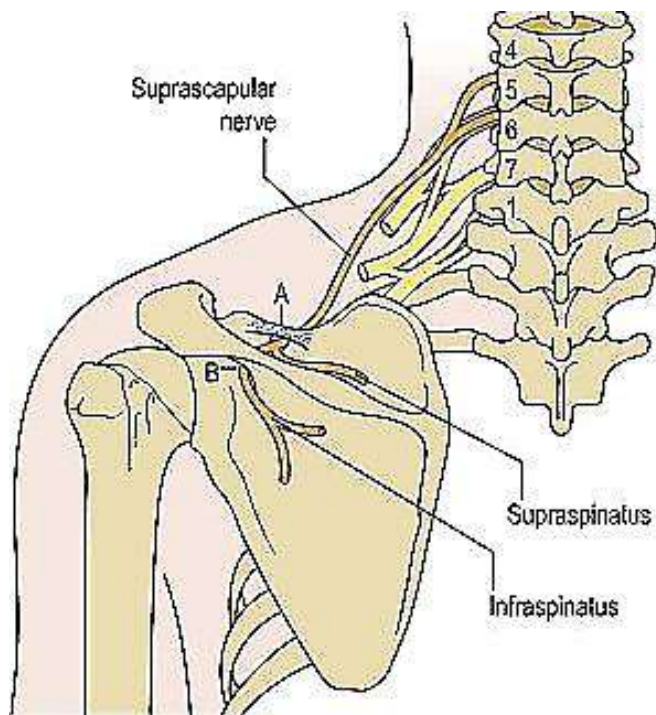
descends superficial to the serratus anterior muscle & supplies it .



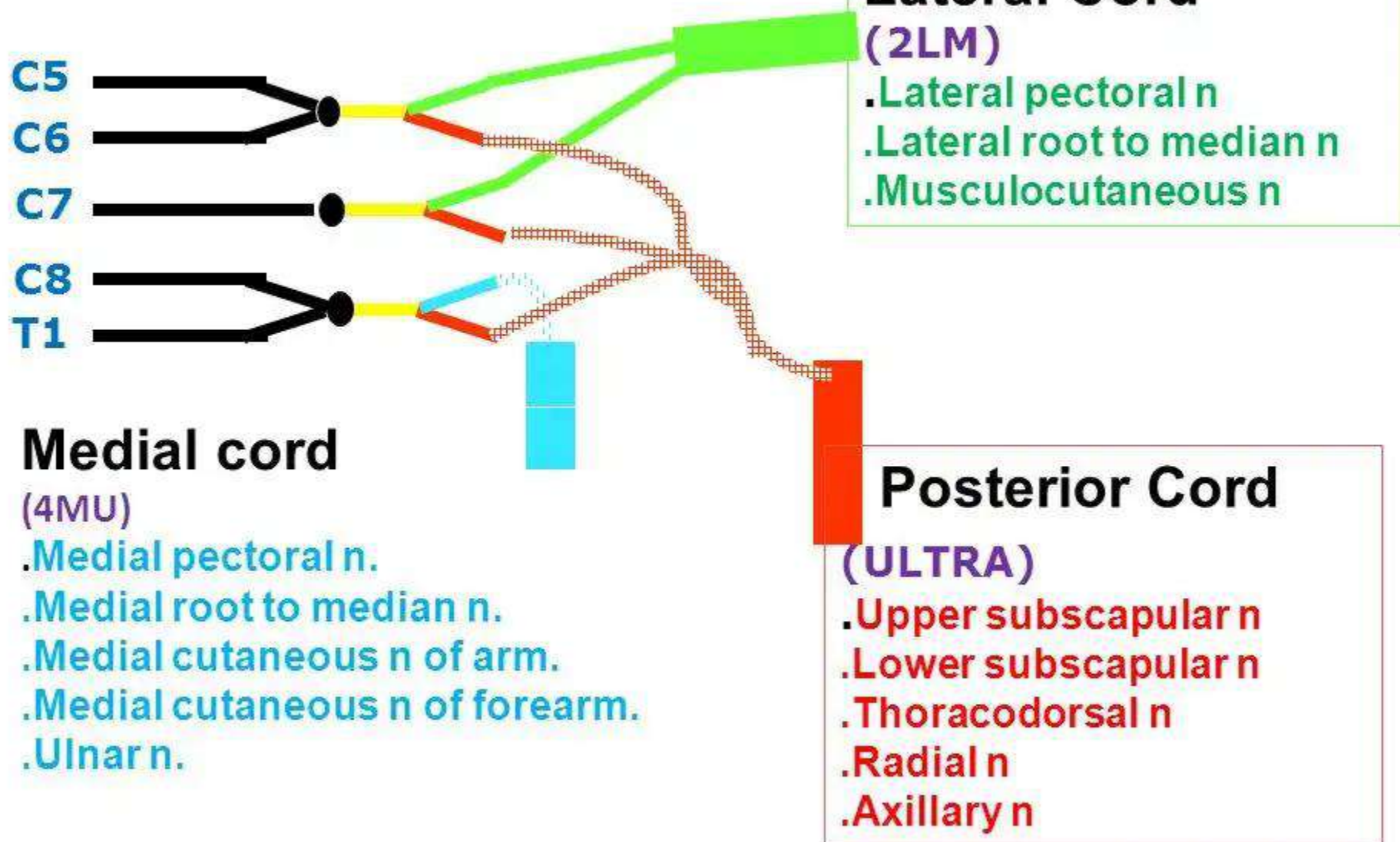
Branches from Trunks

Upper trunk

- **Nerve to subclavius (5,6)** supplies subclavius muscle.
(may give accessory phrenic nerve C5)
- **Suprascapular nerve (C5,6)** supplies supraspinatus and infraspinatus muscles.



C. Branches from Cords



Branches From the Lateral Cord

1. Musculocutaneous nerve (C5,6,7)

..Pierces the coracobrachialis muscle

..Supplies: coracobrachialis, biceps brachii & brachialis.

..It continues as the lateral cutaneous nerve of the forearm

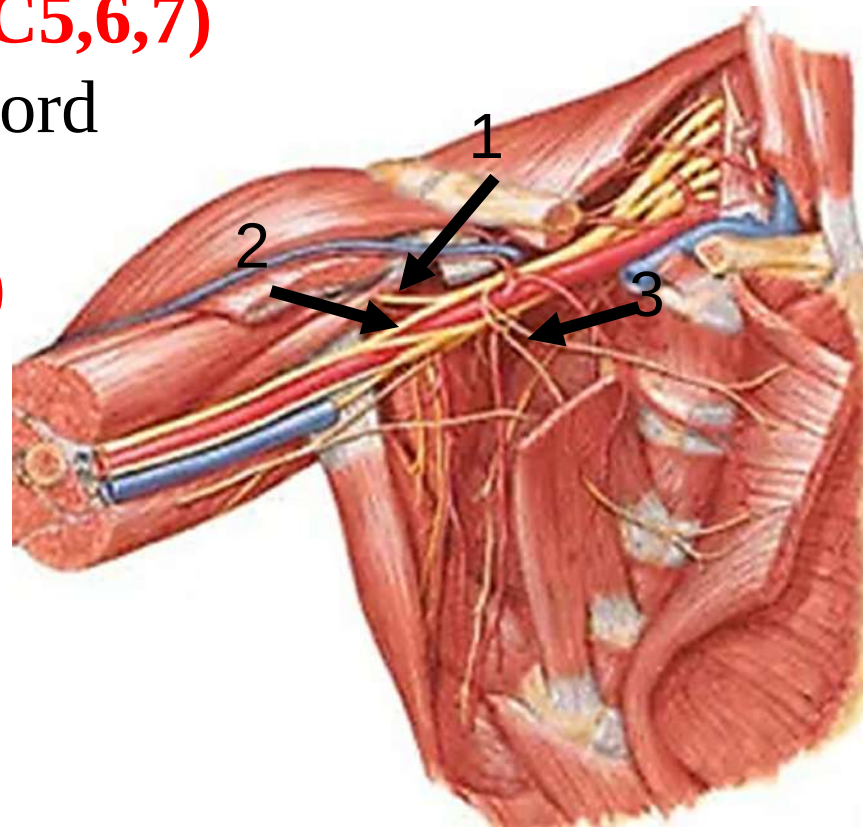
2. Lateral root of median nerve (C5,6,7)

It joins the medial root of medial cord to form the median nerve.

3. Lateral pectoral nerve (C5,6,7)

Pierces the clavipectoral fascia.

Supplies: pectoralis major muscle.



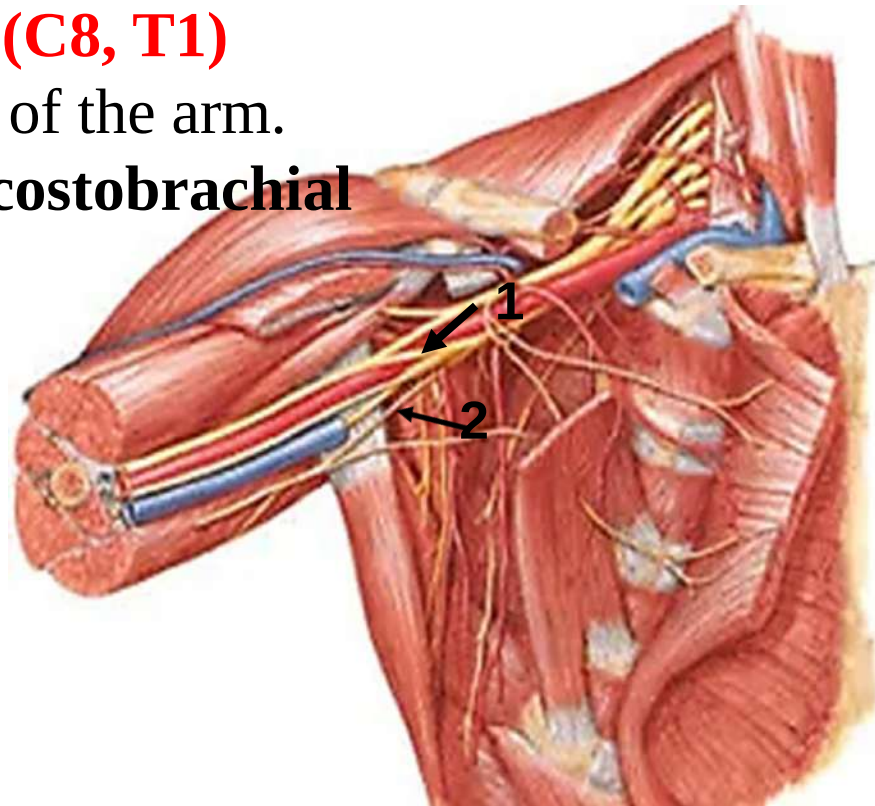
Branches From the Medial Cord

1. Medial root of median nerve (C8, T1)

..Crosses the 3rd part of axillary artery (from medial to lateral) to join the lateral root and forms the median nerve.

2. Medial cutaneous nerve of the arm (C8, T1)

..Supplies the skin of the medial side of the arm.
..It may communicate with **the intercostobrachial nerve (T2).**



Branches From the Medial Cord

3. Medial cutaneous nerve of the forearm (C8, T1)

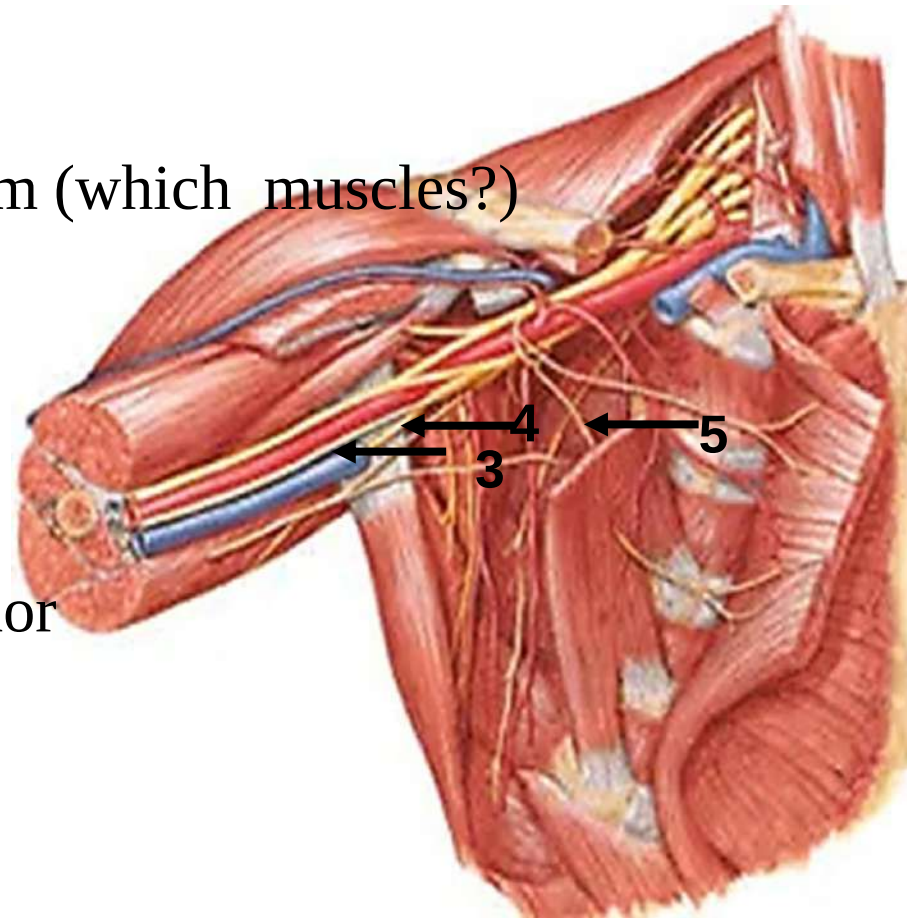
- ..It runs medial to the axillary artery
- ..It supplies the medial side of the forearm.

4. Ulnar nerve (C7,8, T1)

- ..Supplies 1 & ½ muscles of the forearm (which muscles?)
- ..Most muscles of the hand.

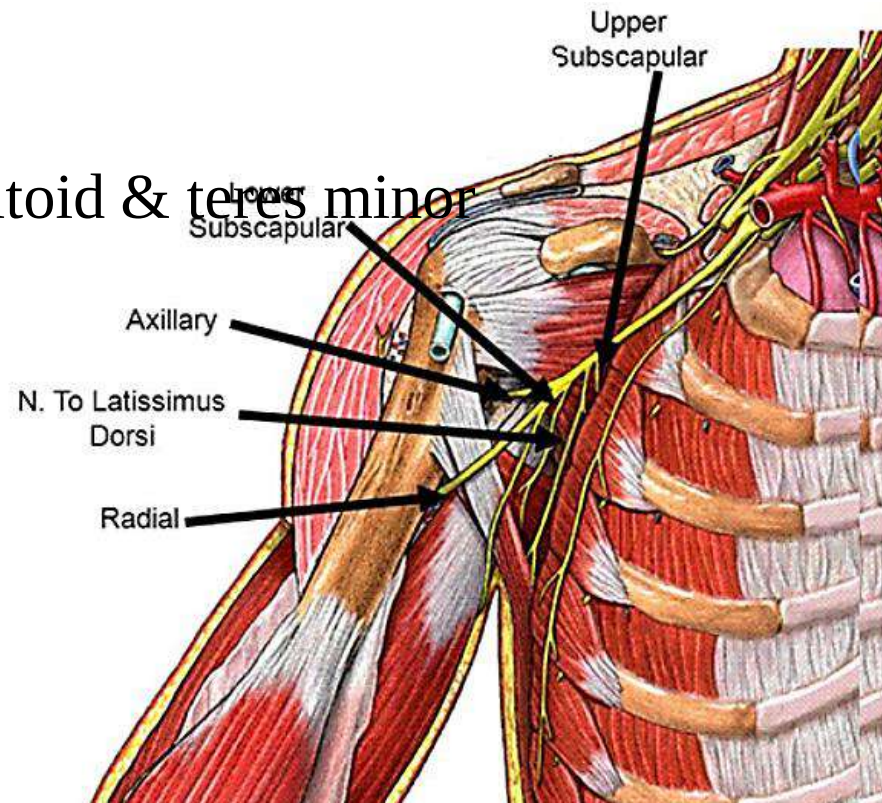
5. Medial pectoral nerve (C8, T1)

- ..Pierces the pectoralis minor muscle
- ..Supplies the pectoralis major and minor

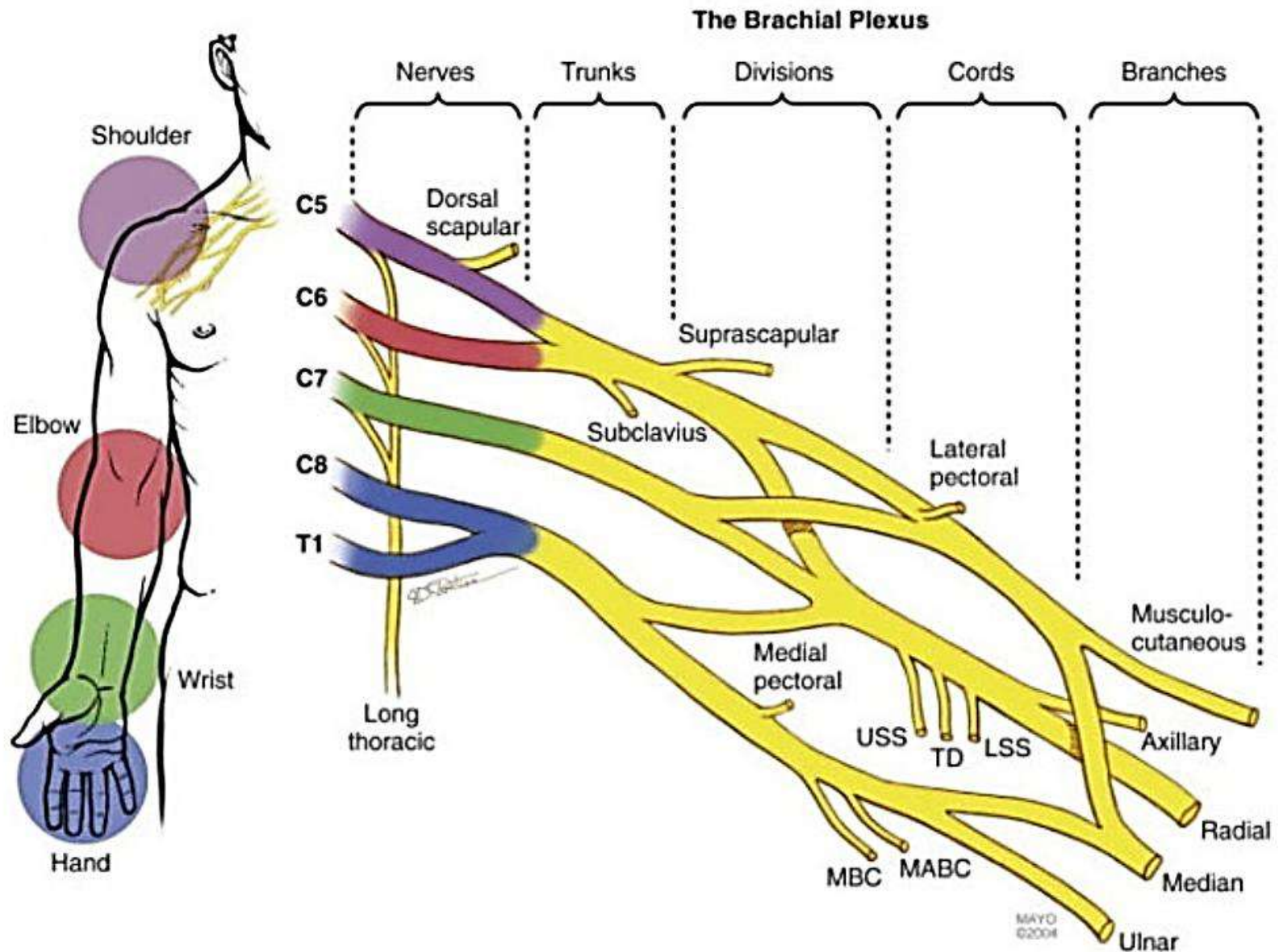


From the Posterior cord

- **Upper subscapular nerve (C5,6)**
supplies the upper part of the subscapularis muscle.
- **Lower subscapular nerve (C5,6)**
supplies: the lower part of the subscapularis and the teres major muscle
- **Thoracodorsal nerve (C6,7,8)**
supplies the latissimus dorsi muscle.
2 terminal branches:
- **Axillary nerve (C5,6)**.: supplies deltoid & teres minor
- **Radial nerve (C5,6,7,8, T1)**



Summary



Clinical correlation

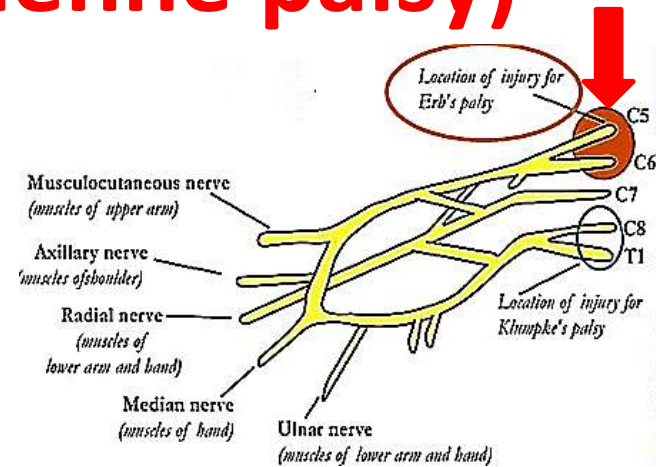
1. Upper trunk lesion(Erb-Duchenne palsy)
2. Lower trunk lesion (klumpke palsy)

1. Upper lesion(Erb-Duchenne palsy)

Causes: traction or tearing of roots C5 C6 as in excessive displacement of the head and depression of the shoulder

Nerves affected:

- ..Suprascapular.
- ..Nerve to subclavius.
- ..Musculocutaneous nerve.
- ..Axillary nerve



Motor affection

Region affected	Muscles paralyzed	Motor action lost
Scapular region & shoulder	Supraspinatus infraspinatus	Abduction Lateral rotation
	Deltoid Teres minor	Abduction Lateral rotation
Ant comp of the arm	Biceps most of Brachialis coracobrachialis	Supination Flexion of elbow Flexion of shoulder

Upper lesion(Erb-Duchenne palsy) cont.

Deformity: porter (waiter's)tip position =

The UL is hanging by the side

is medially rotated

the forearm extended and pronated



Sensory affection

Loss of sensation along the lateral side of the arm

2. Lower lesion (klumpke palsy)

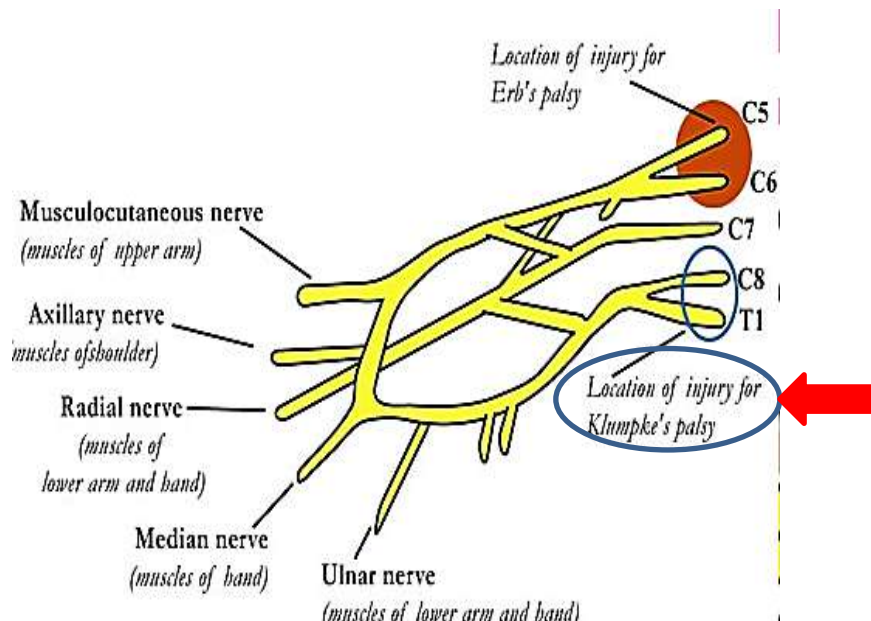
Causes: traction or tearing of root T1 as in.....

- *excessive abduction of the arm,
- *cervical rib
- *metastases of lower deep cervical LN

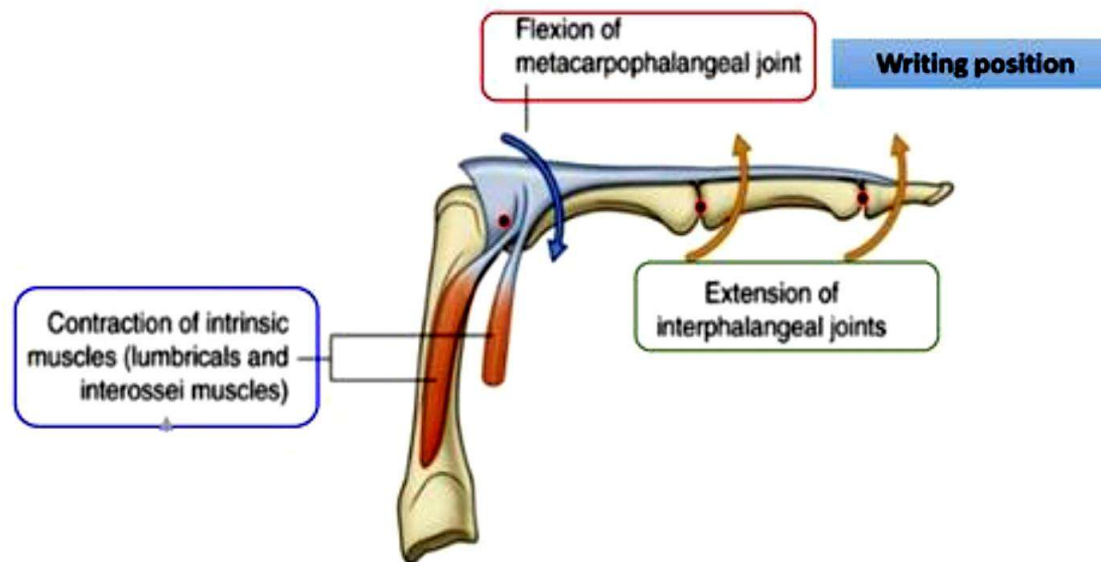
Nerves affected: fibers running in the ulnar & median nerve

Muscles affected: all small muscles of the hand

Motor affection: **lumbricals & interossei**



Action of Lumbricals & Interossei



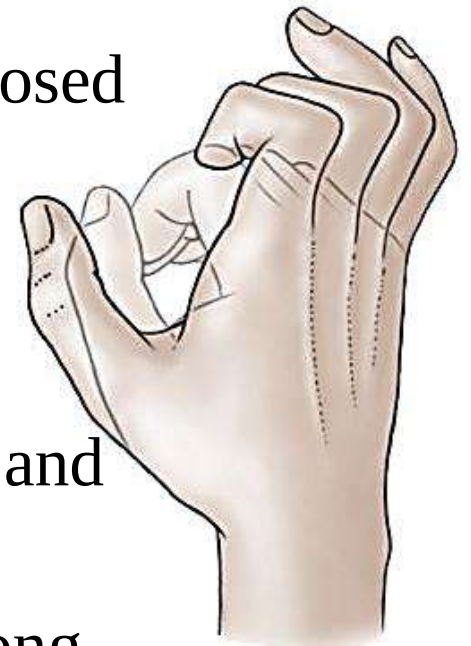
**Paralysis of
lumbricals and
interossei**



Lower lesion (klumpke palsy) cont.

Deformity: claw hand

- hyperextension of the metacarpophalangeal joints (by unopposed extensor digitorum)
- flexion of the interphalangeal joints (by unopposed flexor digitorum sup. & profun)



Sensory affection:

loss of sensation along the medial side of the arm and forearm

N.B.: if C8 is affected sensory loss will extend along the forearm , hand & medial 2 fingers

Lesions of Branches **of Brachial plexus**

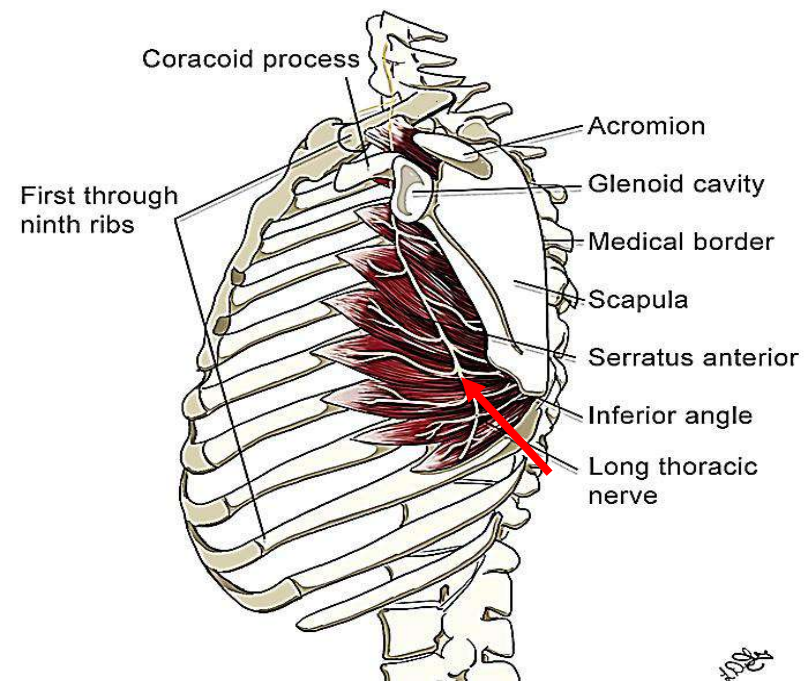
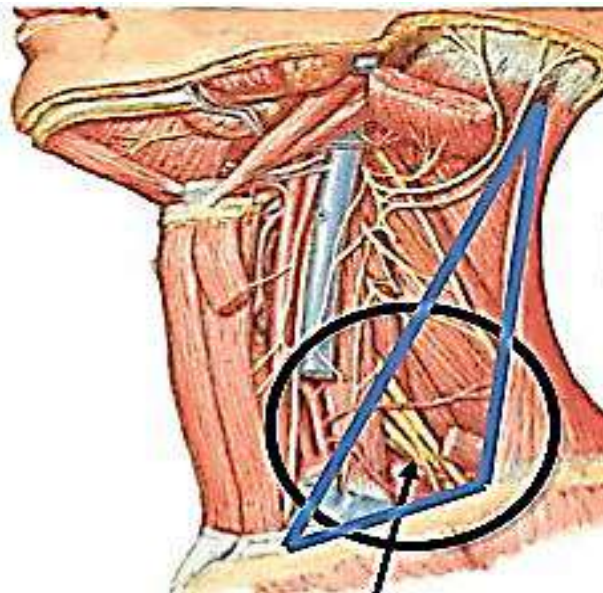
Long Thoracic Nerve

Origin: branch from the roots (C5,6,7)

Causes: affection of the root value C5,6,7

affection of the nerve itself as in cases of:

- lesion in the posterior triangle of the neck
- incision in the medial wall of the axilla.
- during mastectomy



Muscles affected and motor affection:

paralysis of the serratus anterior resulting in...

difficulty in raising the arm above the head (more than 90)

(in ability to rotate of the scapula during abduction)

Deformity : winged scapula

medial border of the scapula will protrude posteriorly



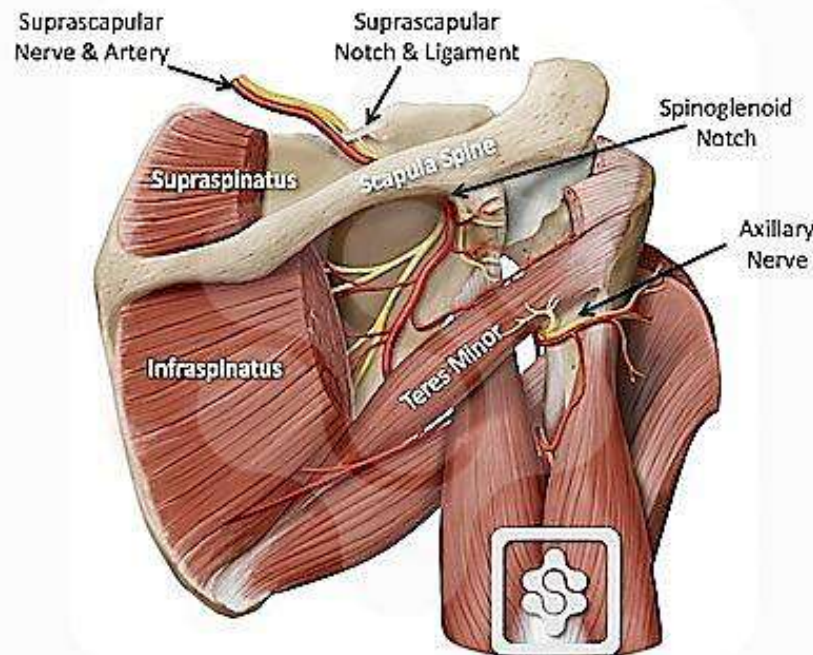
Axillary Nerve

C5, 6

Origin: from the posterior cord of the brachial plexus in the axilla (C5,6).

Course:

- It lies on the subscapularis muscle behind the 3rd part of the axillary artery.
- It enters the quadrangular space with the posterior circumflex humeral artery.
- It divides into anterior and posterior divisions.

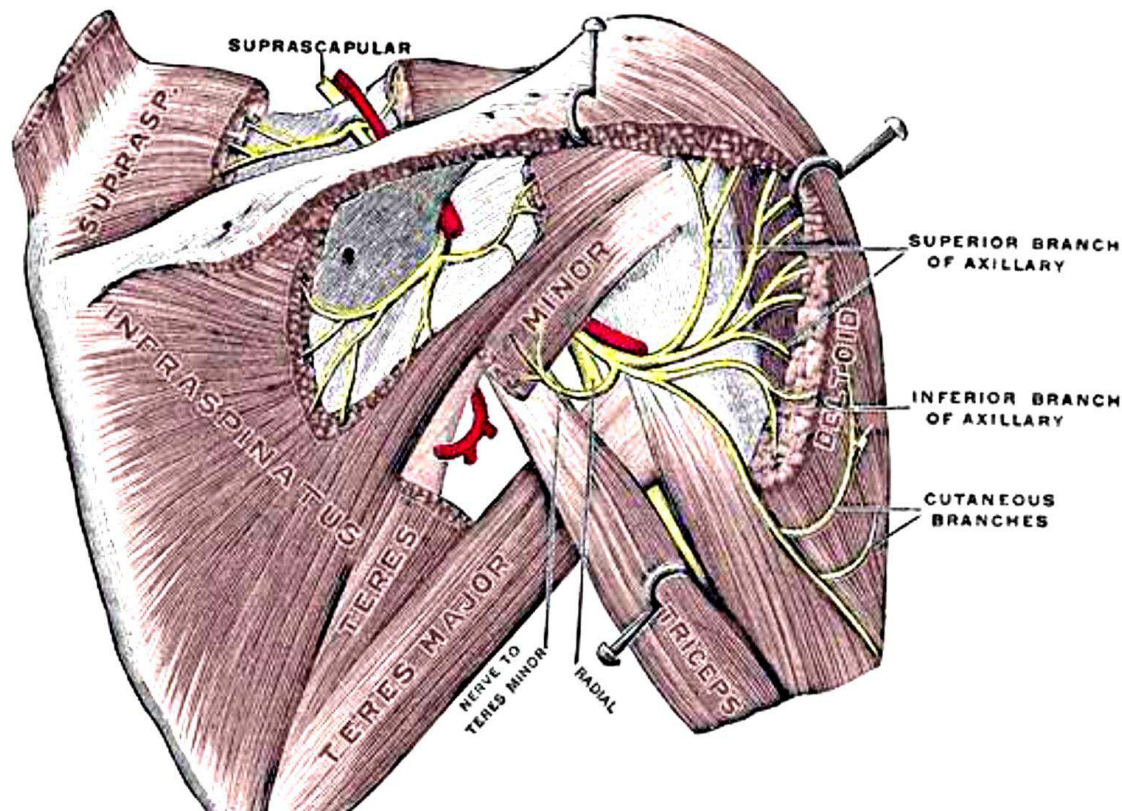


Branches:

Articular branches: to the shoulder joint.

Anterior division: to deltoid and covered its lower part.

Posterior division: to deltoid and teres minor muscles
upper lateral cutaneous nerve of the arm.



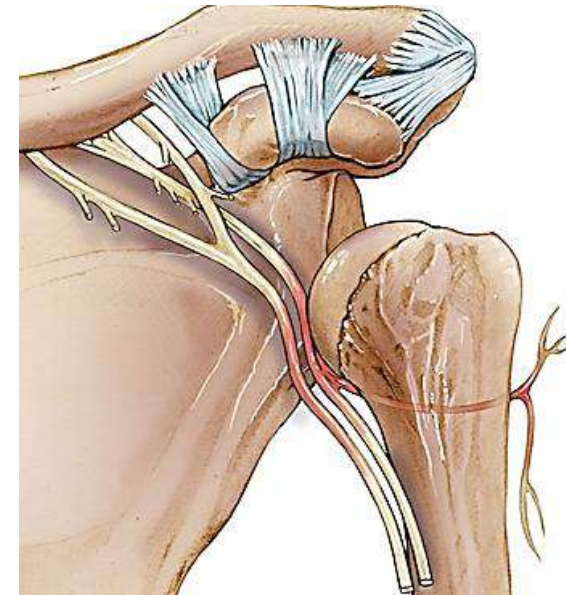
Causes of injury:

- Using the badly fitting crutches
- Inferior dislocation of shoulder
- Fracture of surgical neck of the humerus



Muscles affected & motor affection :

- *Paralysis of deltoid & teres minor
- *Abduction is impaired



Deformity:

Wasting of the deltoid muscle leading to flat shoulder
Impairment of abduction of the shoulder

Sensory affection:

loss of sensation along the **lower half** of the deltoid muscle



Radial Nerve

C5,6,7,8,T1

Origin: from the posterior cord of the brachial plexus in the axilla (C5,6,7,8,T1)

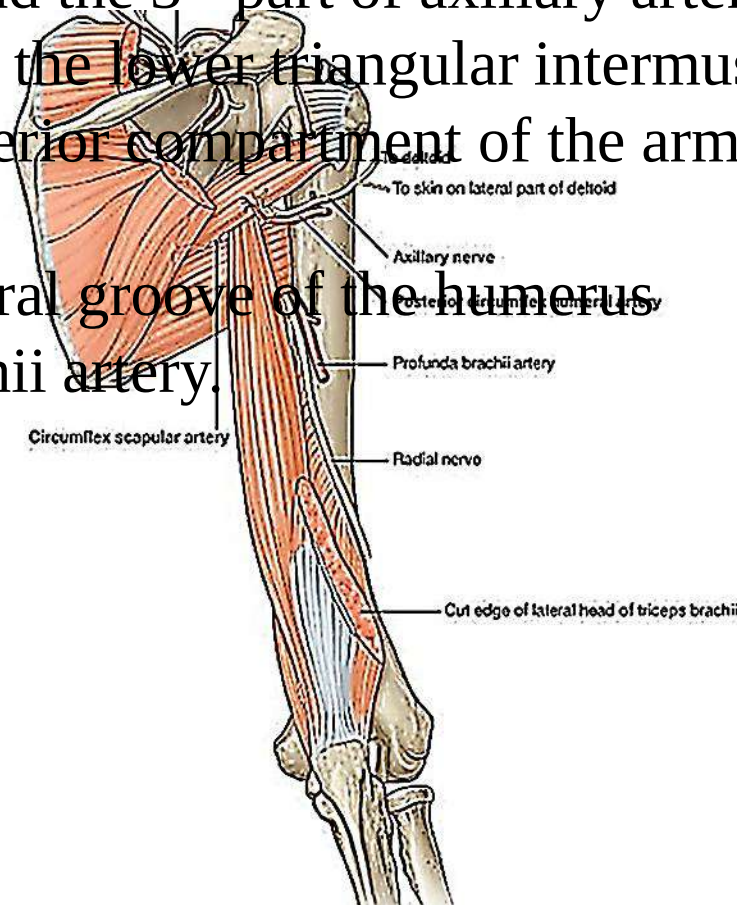
Course:

Axillary course

- It descends behind the 3rd part of axillary artery.
- It passes through the lower triangular intermuscular space.
- It enters the posterior compartment of the arm.

Spiral groove

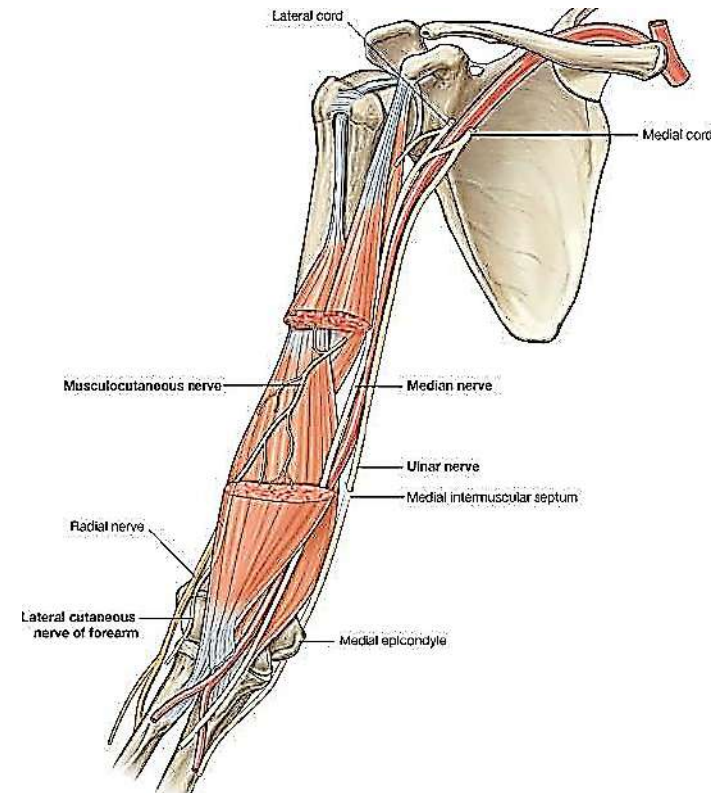
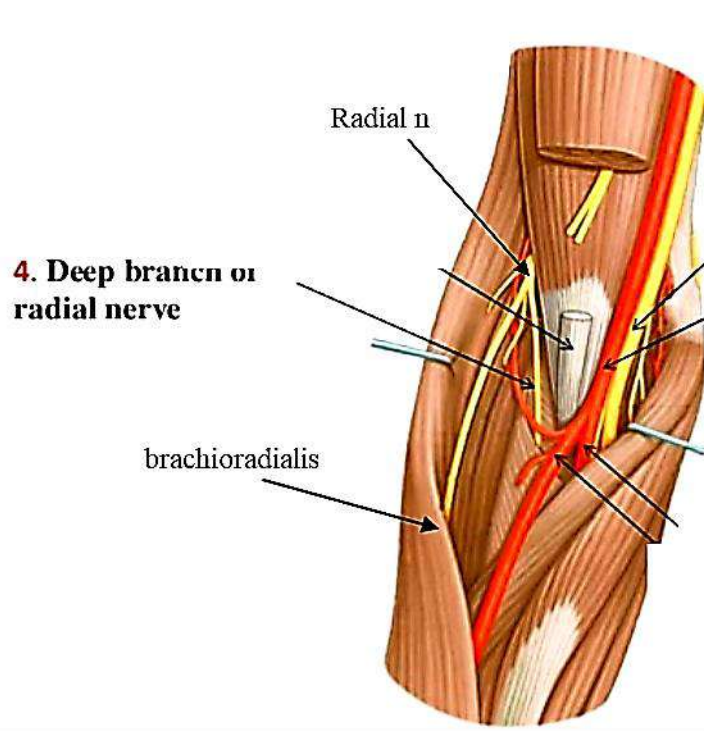
- It runs on the spiral groove of the humerus with profunda brachii artery.



- It emerges between brachialis and brachioradialis in the anterior compartment of the arm.

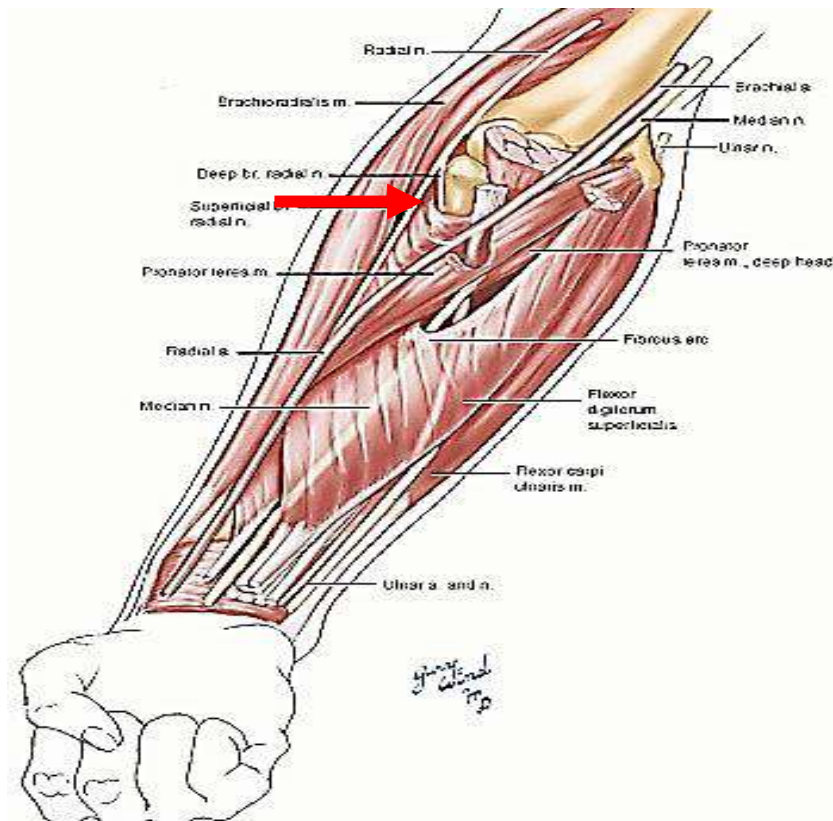
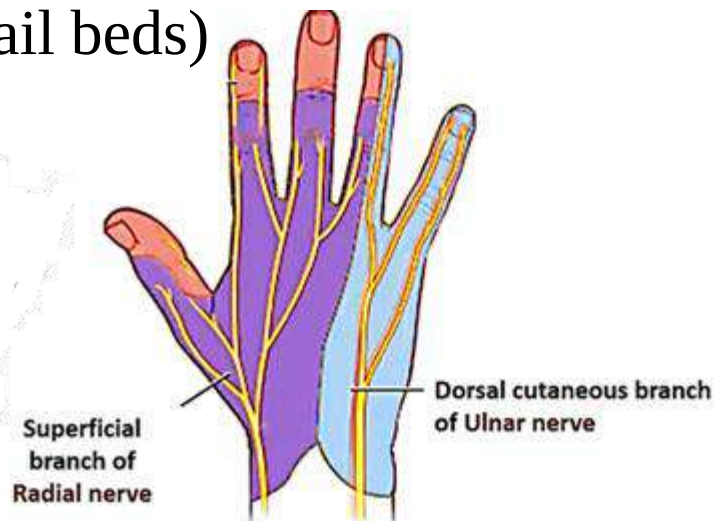
Forearm division

- It passes in front of the lateral epicondyle of the humerus to enter the forearm.
- It divides into its 2 (superficial and deep) terminal branches in front of the lateral epicondyle of the humerus.



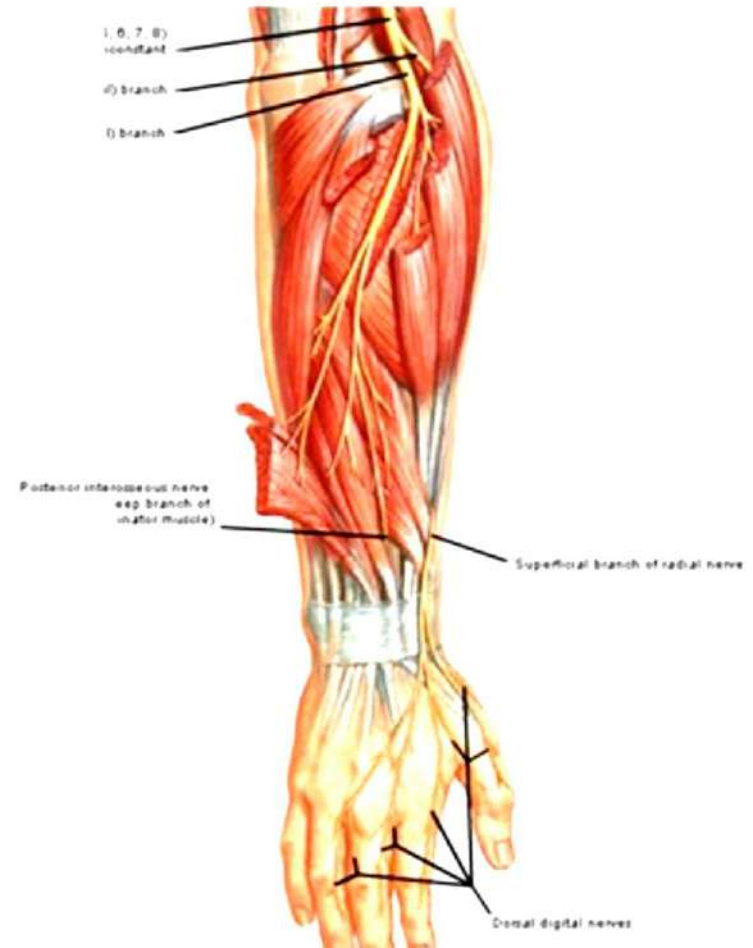
The superficial branch of the radial nerve

- It runs under the brachioradialis muscle, on the lateral aspect of the radial artery.
- In the distal part of the forearm, it passes backward under the tendon of brachioradialis to reach the posterior surface of the hand.
- It supplies the skin of :
lateral 2/3 of the dorsum of the hand &
dorsal surface of the lateral 3&1/2
(except the nail beds)



The deep branch of the radial nerve

- It pierces the supinator muscle, and winds around the neck of the radius (inside the supinator).
- It leaves the supinator (as the posterior interosseous nerve) and runs between the superficial and deep extensor muscles in the posterior compartment of the forearm.



Branches of the Radial nerve

1. In the axilla:

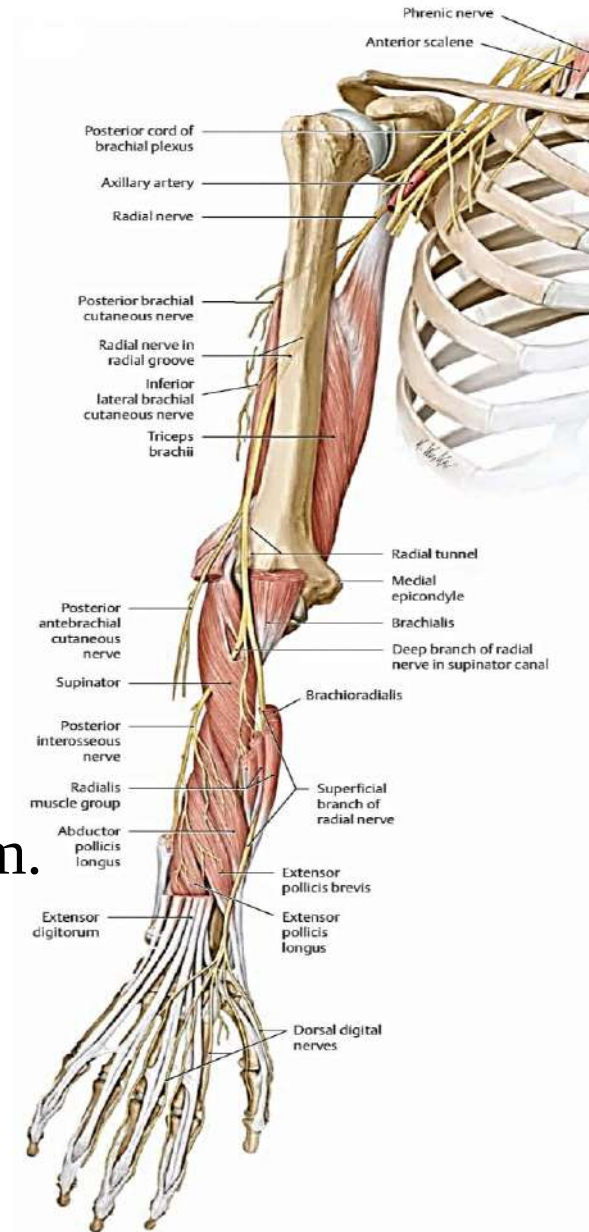
Muscular: 1. long head of triceps.
2. medial head of triceps.

Sensory: 3. Posterior cutaneous nerve of the arm.

2. In the spiral groove:

Muscular: 1. lateral head of triceps.
2. medial head of triceps.
3. anconeus muscle.

Sensory: 4. Lower lateral cutaneous nerve of the arm.
5. Cutaneous nerve of the forearm.

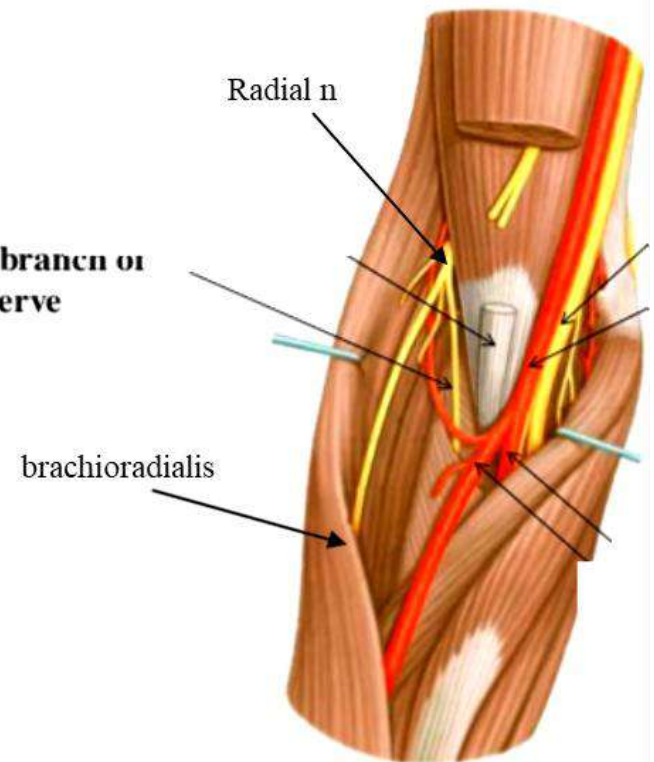


3. In the (ant comp) groove between brachialis and brachioradialis:

Muscular: 1. lateral part of brachialis.
2. brachioradialis.
3. extensor carpi radialis longus.

Articular: 4. to the elbow joint.

4. Deep branch of radial nerve



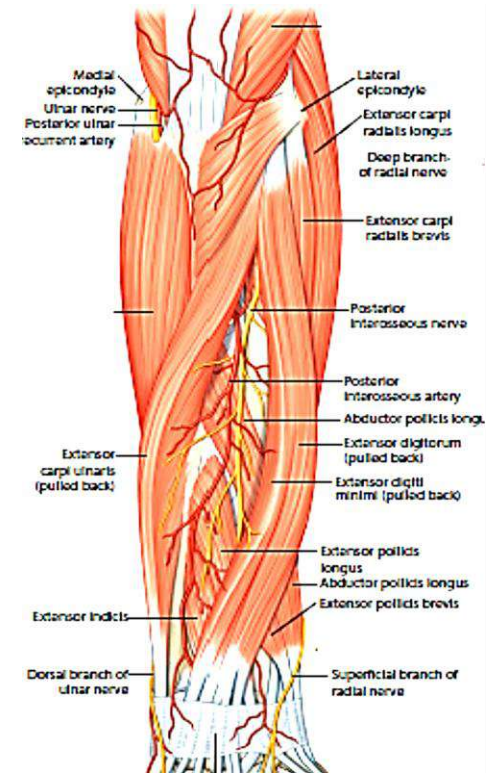
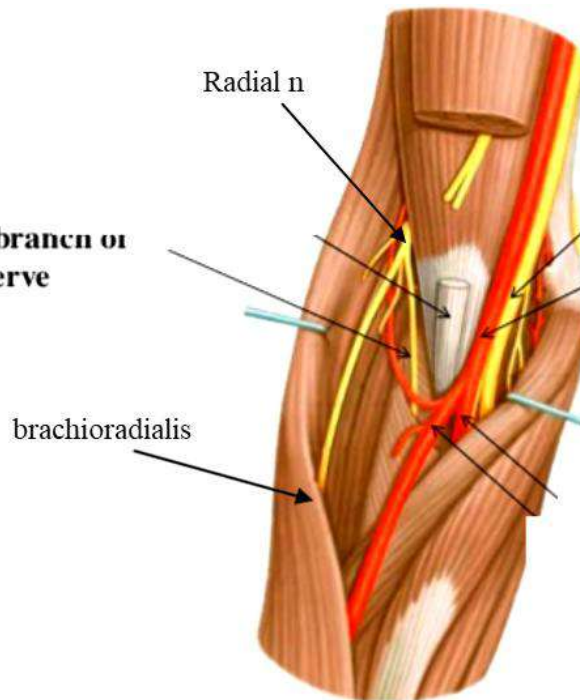
4. In the cubital fossa (anterior to the lateral epicondyle)

Deep branch of radial N: supplies supinator, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi, extensor carpi ulnaris

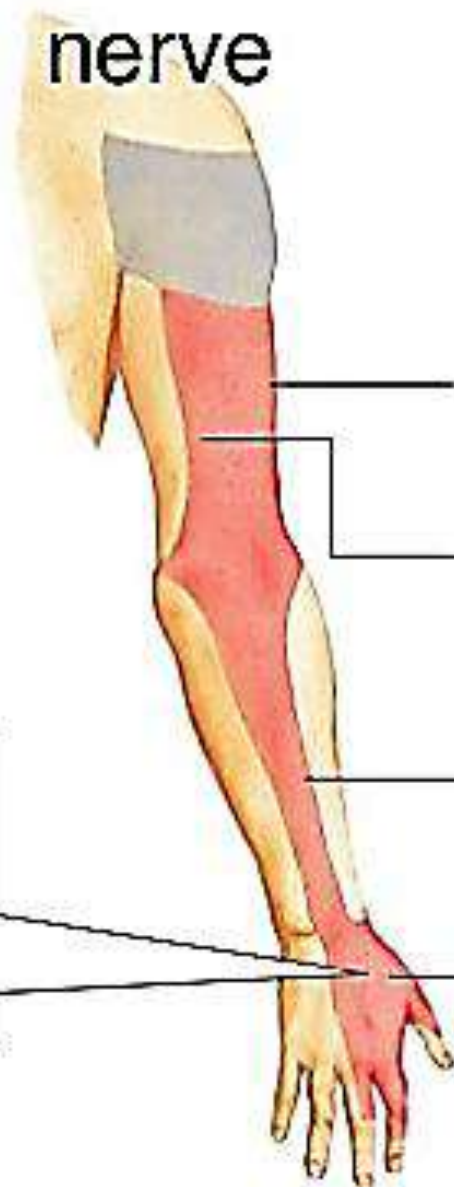
Posterior interosseous (after it emerges from supinator muscle):

- **remaining muscles** of the posterior compartment of the forearm: abductor pollicis long., extensor polli. brevis, extensor polli. longus and extensor indicis.
- articular brs to the wrist & carpal joints.

4. Deep branch of radial nerve



Cutaneous innervation from radial nerve



Lower lateral cutaneous nerve of arm

Posterior cutaneous nerve of arm

Posterior cutaneous nerve of forearm

Dorsal radial sensory nerve

Gives sensibility to the **dorsum** of the hand over the **radial** two-thirds, the **dorsum** of the thumb, **and** the index, **middle** finger **proximal** to the **distal** interphalangeal joint.

1-Lesion of radial nerve in the axilla

Causes: badly fitting crutches

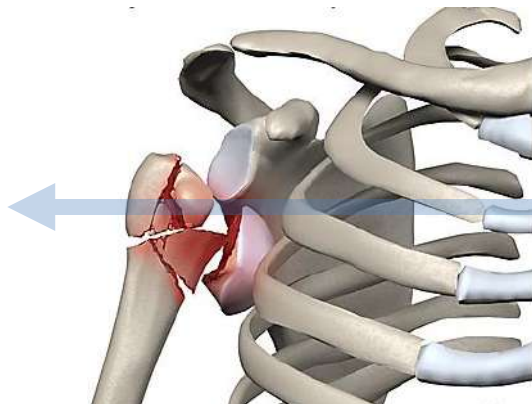
Saturday night palsy

dislocated fracture of proximal part of the humerus

Muscles affected:

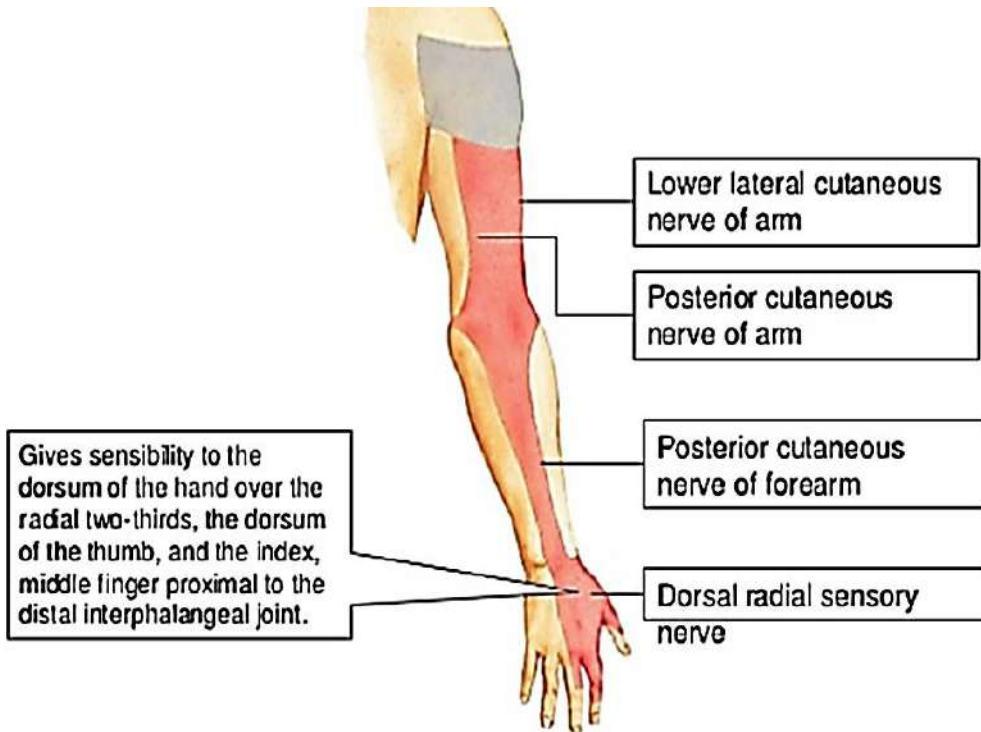
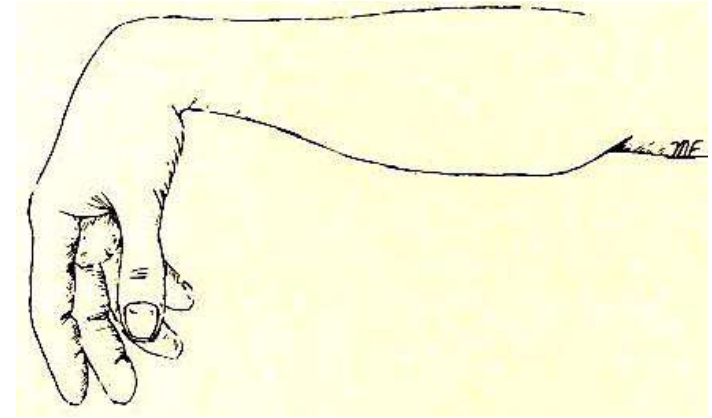
Triceps, anconeus, long extensors of the wrist

Motor affection: inability to extend the elbow, wrist & fingers



Deformity: wrist drop

by the unopposed action of the flexors of the wrist & fingers



Sensory loss : of the posterior surface of the back of the arm, the back of the forearm, lateral 2/3 of the dorsum of the hand and lateral 3.5 finger (except nail beds)

2- Lesion of radial nerve in the spiral groove

Causes: Prolonged application of tourniquet
fracture of mid shaft of the humerus
pressure of the back of the arm on the edge operating table

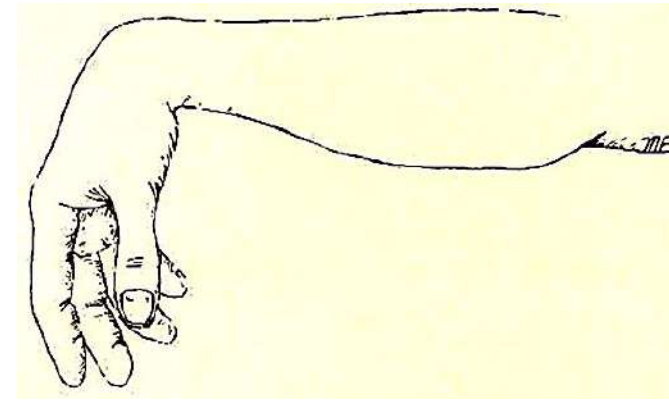
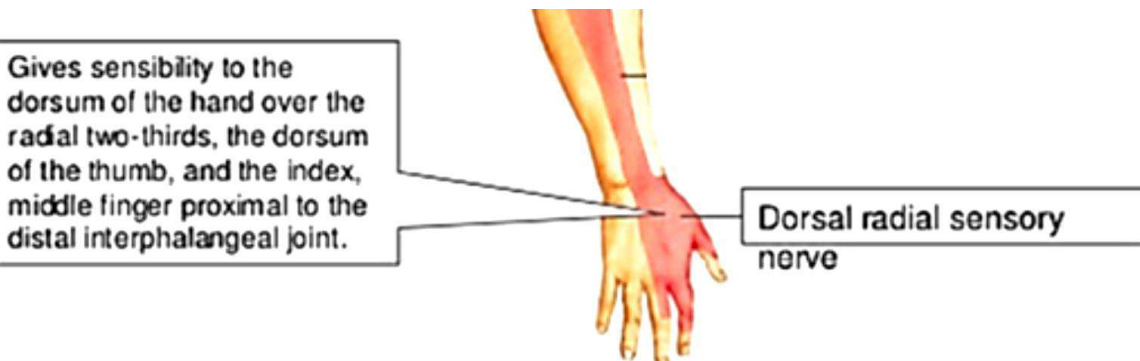
Muscles affected: long extensors of the wrist & fingers

Motor affection: inability to extend wrist & fingers

Deformity: wrist drop

by the unopposed action of the flexors of the wrist & fingers

Sensory: the lateral 2/3 of the dorsum of the hand and lateral 3.5 finger (except nail beds).



3- Lesion of the deep br. of radial nerve

Causes:

- fracture of the proximal end of the radius
- fracture of the distal end of the humerus
- dislocation of the elbow



Muscles affected: long extensors of the wrist & fingers (except the extensor carpi radialis longus and supinator which are supplied by the radial nerve)



Deformity: No wrist drop: as the extensor carpi radialis longus can extend the wrist

Sensory affection: No sensory loss



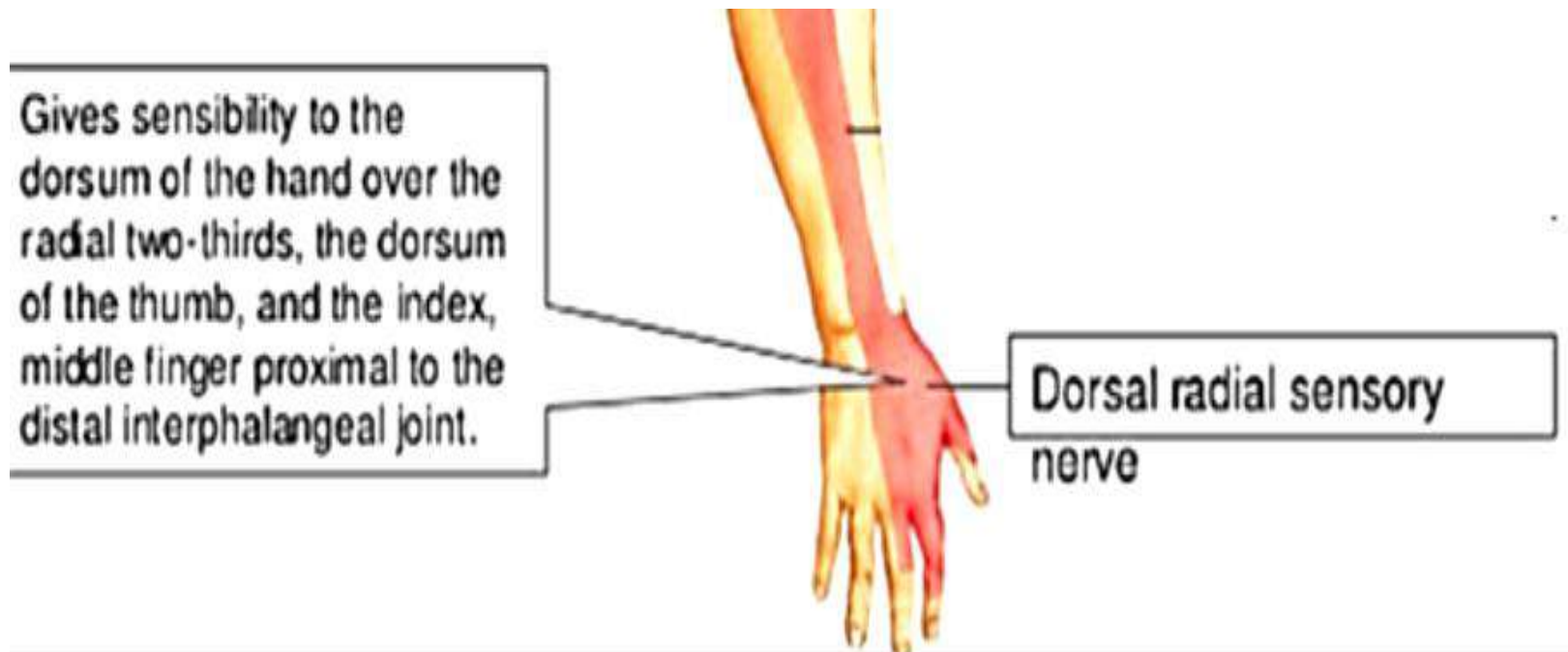
4- Lesion of the superficial branch of the radial nerve

Causes:

Direct stab wound

Motor affection: No motor affection

Sensory loss of the lateral 2/3 of the dorsum of the hand and lateral 3.5 finger (except nail beds).

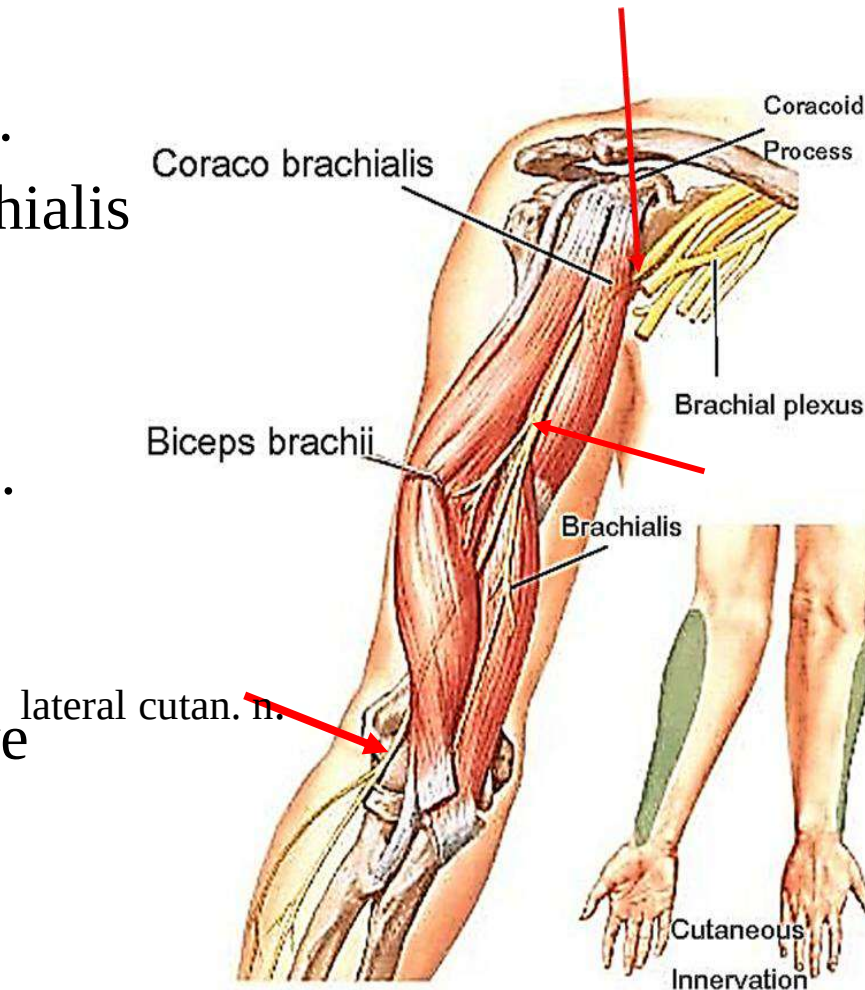


Musculocutaneous nerve
C5,6,7

Origin: from lateral cord of the brachial plexus in the axilla (C5,6,7).

Course:

- It pierces the coracobrachialis muscle.
- It passes between the biceps and brachialis
- It pierce the deep fascia at the lateral margin of biceps to becomes superficial.
- It descends on the lateral side of the forearm as the lateral cutaneous nerve of the forearm.



Branches:

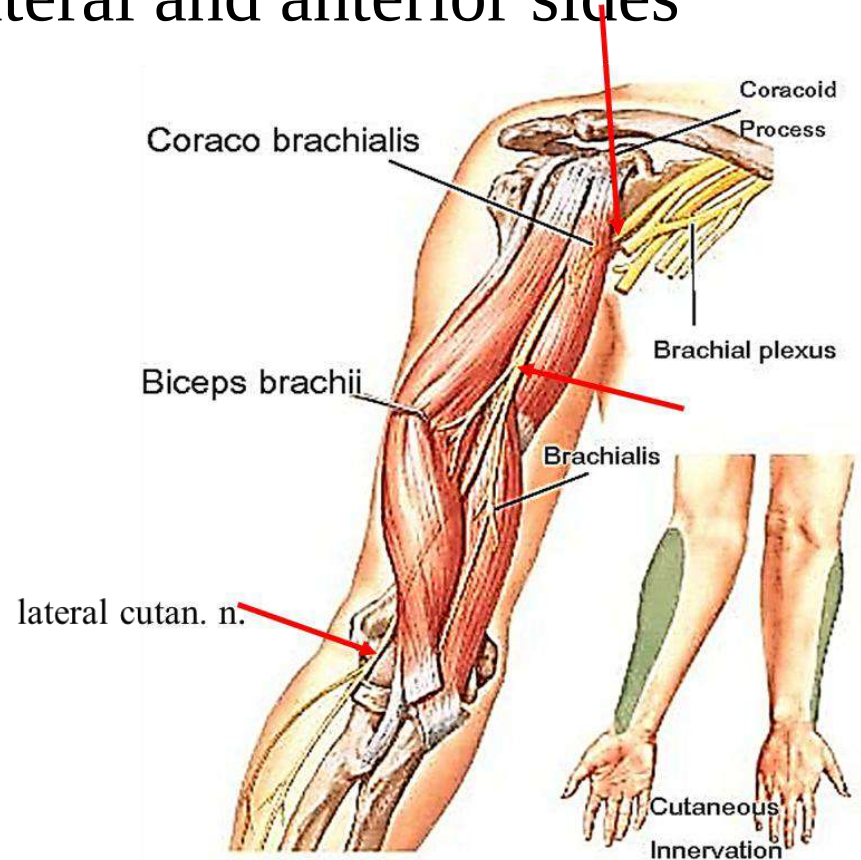
Muscular branches: to coracobrachialis, biceps and majority of brachialis.

Articular branches: to the elbow joint.

Cutaneous branches: to the lateral and anterior sides of the forearm.

Lesion of the nerve

1. High up in the arm
2. In the forearm



1- High up in the arm (This rare)

Muscles affected:

Biceps, coracobrachialis

Medial half of the brachialis

Motor affection:

Flexion of the elbow is weakened and can still be done partly

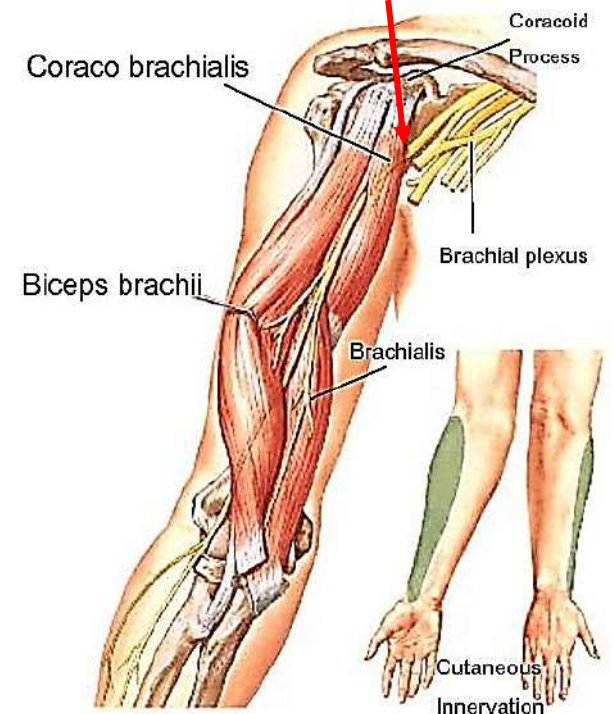
by the : remainder of brachialis, muscles attached to medial epicondyle.

& by brachioradialis and extensor carpi radialis longus in prone position

Sensory affection: along the lateral side of the forearm

2- In the forearm

Sensory affection: along the lateral side of the forearm

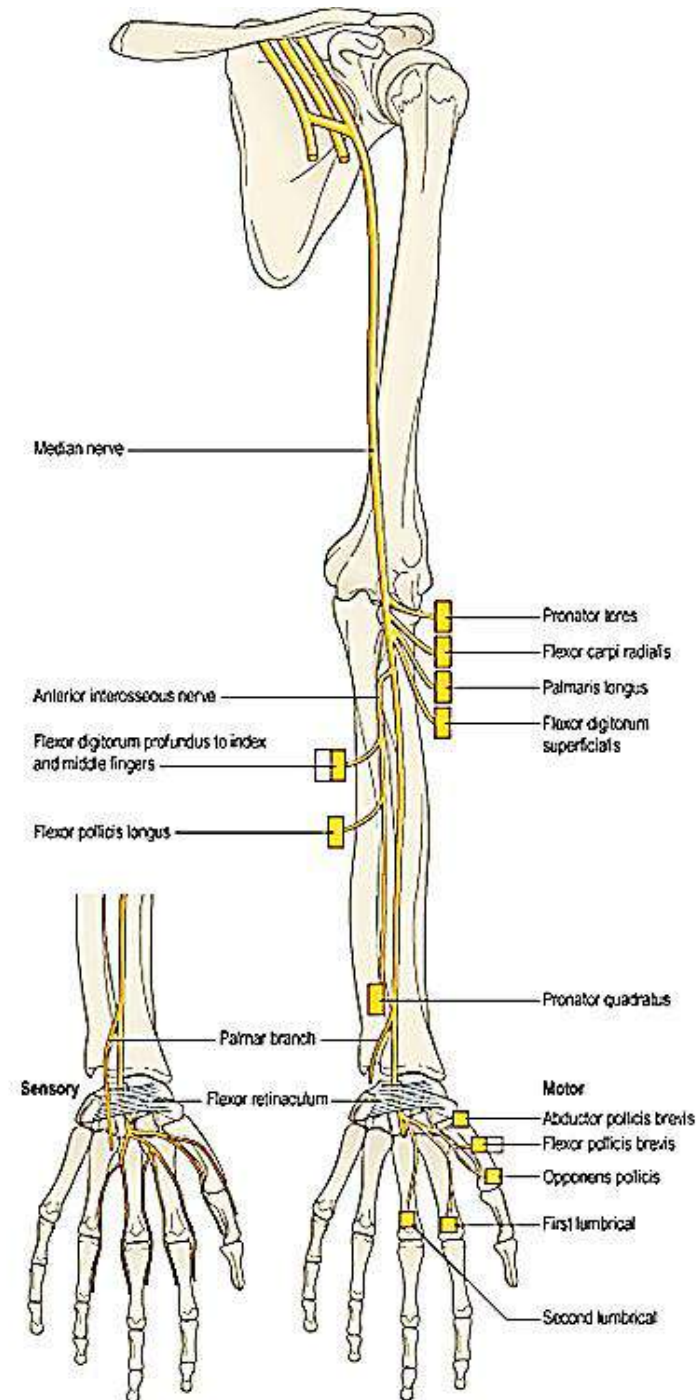


Median Nerve
C5,6,7,8,T1

Median nerve:(C5,6,7,8,T1)

Origin:

- Medial root from the medial cord of brachial plexus (C8, T1)
- Lateral root from the lateral cord of the brachial plexus in the axilla (C5,6,7)

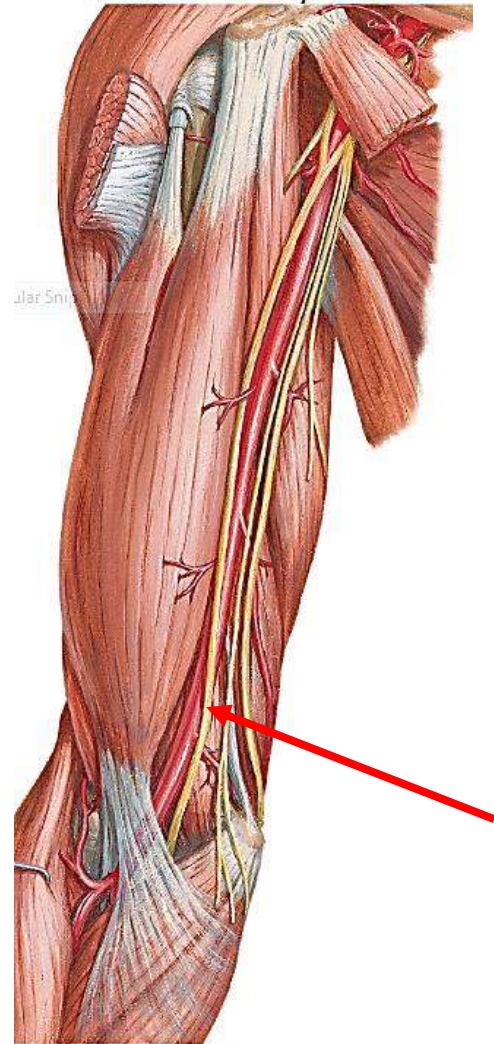


Course in the axilla and arm:

- The medial root crosses the 3rd part of the axillary artery to join the lateral root forming the median nerve (in the axilla).
- It descends on the lateral side of the brachial artery till the middle of the arm.
- It crosses the artery (from lateral to medial) and continues on its medial side.
- At its lower part, it is crossed by the bicipital aponeurosis.
- It continues in the cubital fossa.

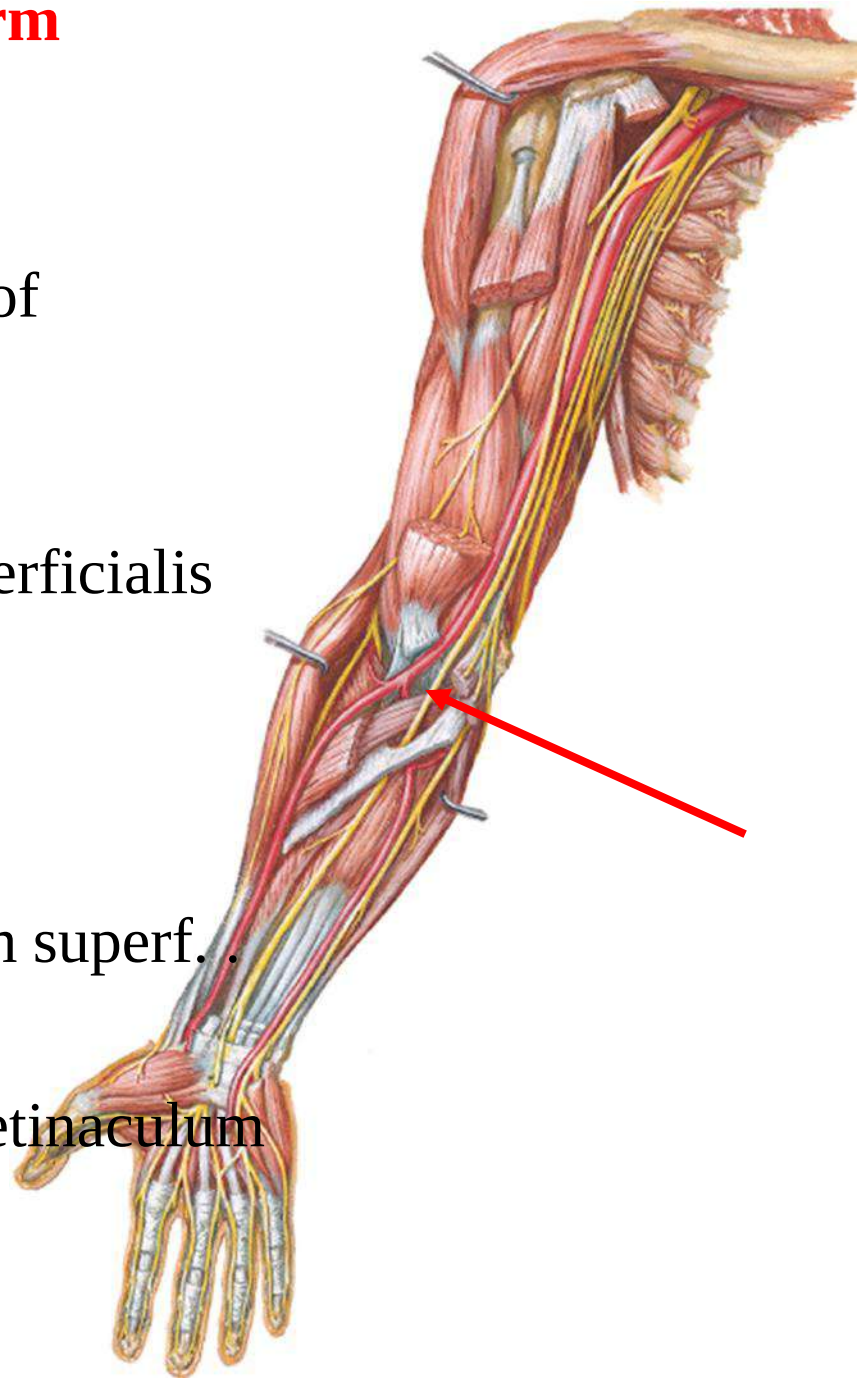
Branches:

No branches in the axilla and arm.



Course in the cubital fossa and forearm

- In the cubital fossa, it runs on the medial side of the brachial artery.
- It enters forearm between the 2 head of pronator teres.
- It runs between flexor digitorum superficialis and profundus.
- Its lower part, becomes superficial on the lateral side of the flexor digitorum superf. .
- It enters the hand deep to the flexor retinaculum



Branches

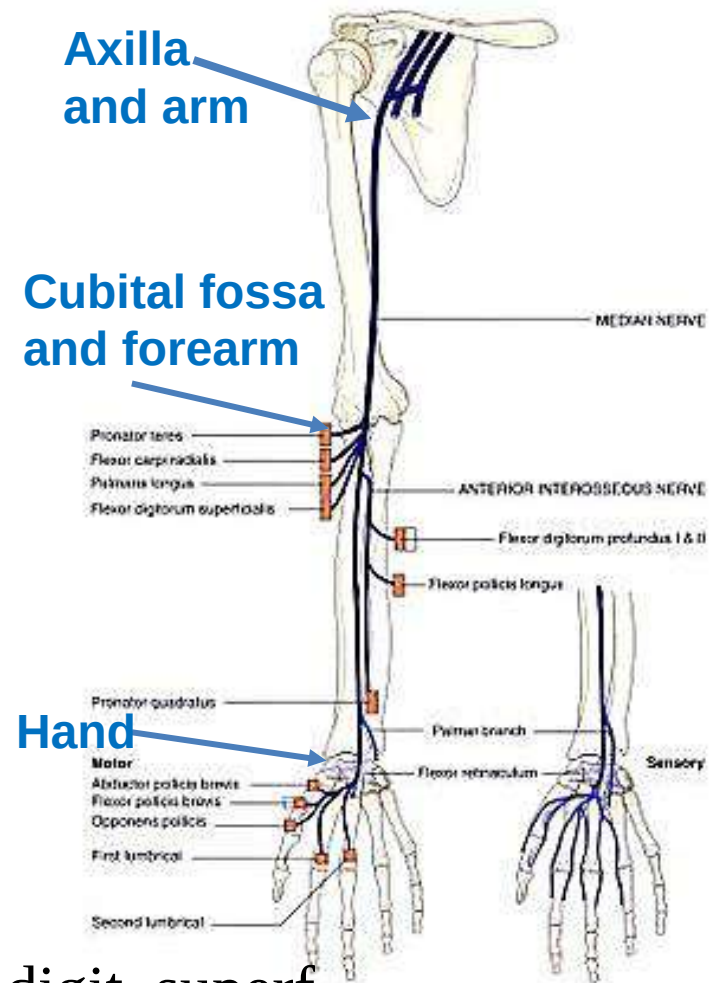
In Axilla and Arm

No branches

In Cubital Fossa and Forearm

Branches:

1. **Muscular branches:** to pronator teres, flexor carpi rad., palmaris longus and flexor digit. superf..
2. **Articular branches:** to the elbow joint.
3. **Palmar cutaneous branches:** passes superficial to the flexor retinaculum and supplies skin of the lateral 2/3 of the palm.
4. **Anterior interosseus nerve**.....



4. Anterior interosseus nerve.:

Origin: from the median nerve as it emerges between the 2 heads of the pronator teres.

Course:

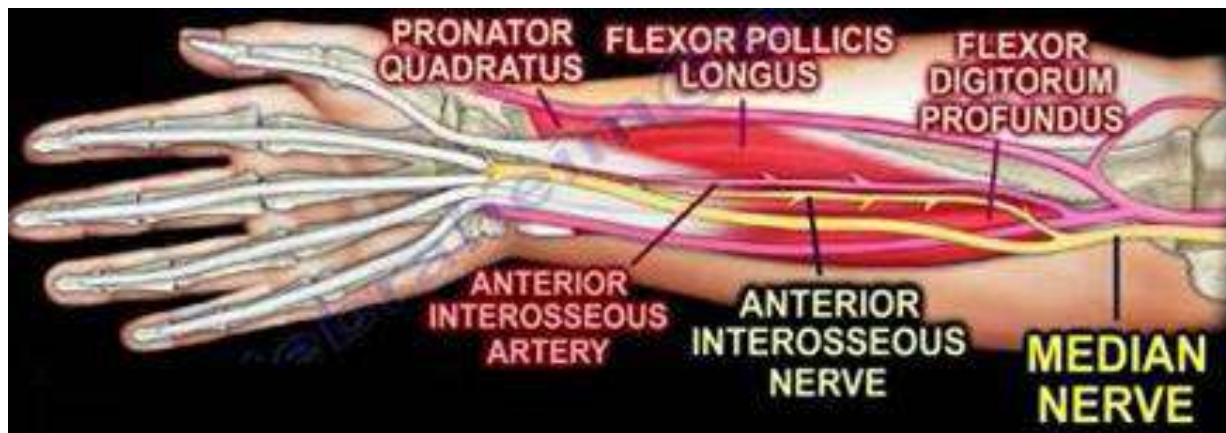
It descends on the interosseus membrane (with the anterior interosseus art.) between the flexor pollicis longus and flexor digitorum profundus.

End: It ends on the anterior surface of the carpal bones.

Branches:

Muscular: to flexor pollicis longus, pronator quadratus and the lateral ½ of flexor digit. prof..

Articular: to wrist, inferior radio-ulnar and intercarpal joints.



In the Hand

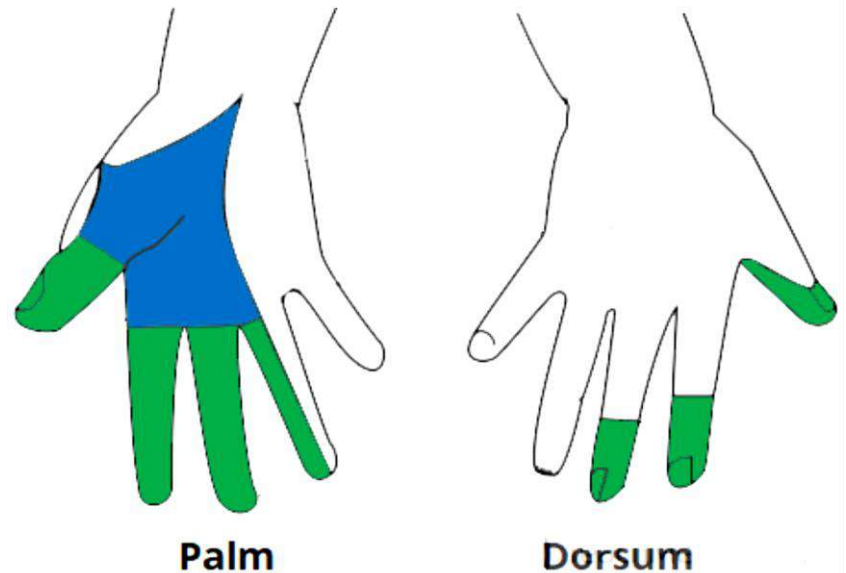
It enters the hand: deep to the flexor retinaculum (inside the carpal tunnel) deep to palmar aponeurosis.

Branches

- **Muscular branches:** to thenar muscle (abductor pollicis brevis, flexor pollicis brevis and opponens pollicis) and the lateral 2 lumbricals.
- **Cutaneous branches:** to the palmar surface of the lateral 3.5 fingers including their nail beds.



- Palmar cutaneous branch
- Digital cutaneous branch



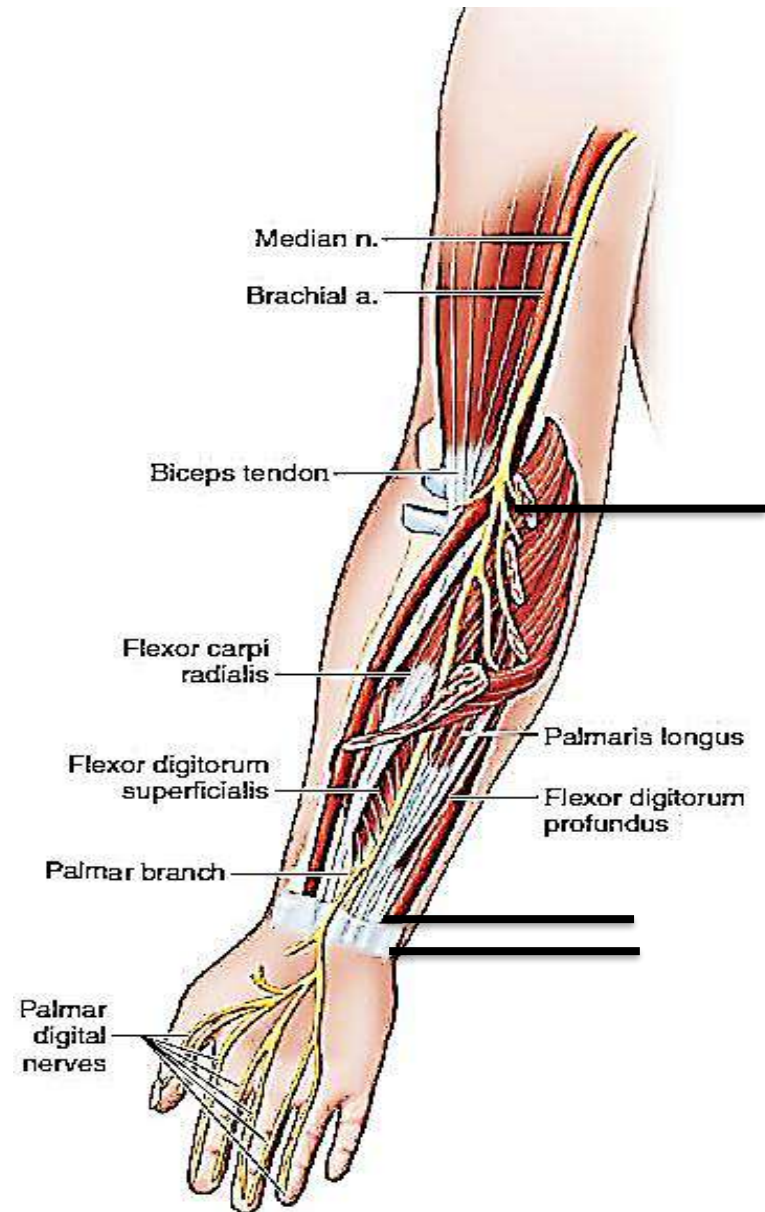
Median Nerve

Injury of the nerve may occur :

1- at the elbow

2- at the wrist

3- carpal tunnel syndrome



1. Lesion at the elbow

Causes:

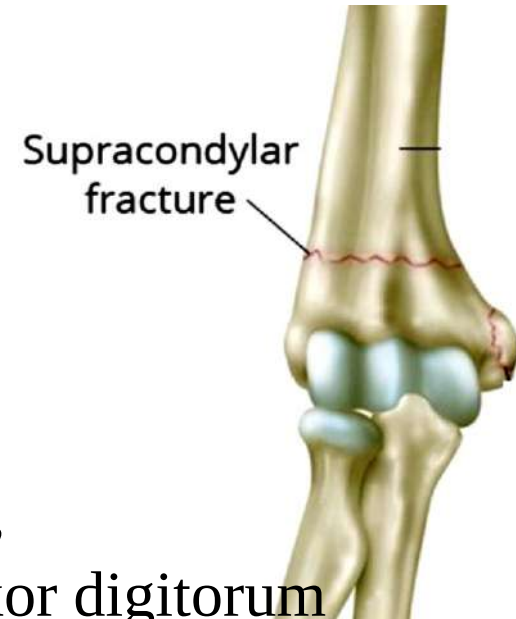
Supracondylar fracture of the humerus.

Muscles affected

- pronator muscles
- all flexors of the wrist (except flexor carpi ulnaris),
- all flexors of the fingers (except medial half of flexor digitorum profundus).

Motor affection

- the forearm supinated
- the hand extended and adducted.
- Extension of the thumb, index and middle fingers,
- but weak flexion of ring and little fingers can occur.
- Paralysis and wasting of thenar muscles.



Deformity: *“Ape-like hand” deformity.*

due atrophy of thenar muscles and inability of the thumb to perform opposition

Sensory loss: of lateral 2/3 of the palm of the hand and anterior surface of the lateral 3&1/2 finger (**including their nail beds**).

Vasomotor and trophic changes.



Ape hand
Deformity



2. Lesion at the wrist

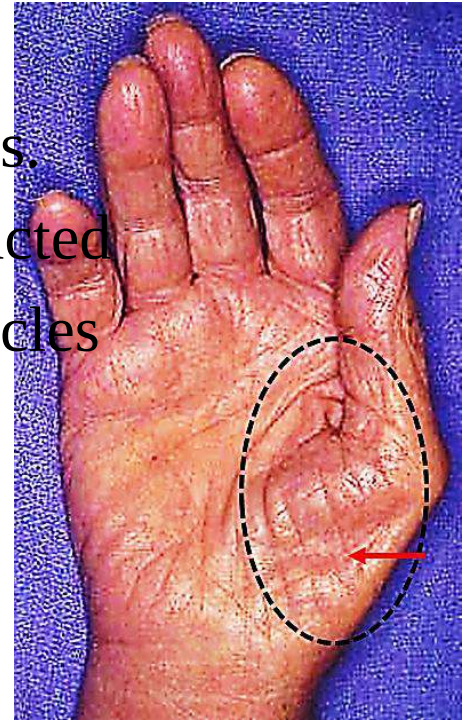
Causes:

Stab wound just above the flexor retinaculum.

Muscles affected: thenar and lateral 2 lumbrical muscles.

Motor affection: the thumb is laterally rotated and adducted and can not oppose other fingers. Wasting of thenar muscles

Deformity: giving *“Ape-like hand” deformity*.



Sensory loss: of, lateral 2/3 of the palm of the hand and anterior surface of the lateral 3 & 1/2 finger (including their nail beds).

Vasomotor and trophic changes.

3.Carpal Tunnel Syndrome

Causes:

Compression on the median nerve inside the carpal tunnel due to thickening of the synovial sheath of the flexor tendons or arthritis of the carpal bones.

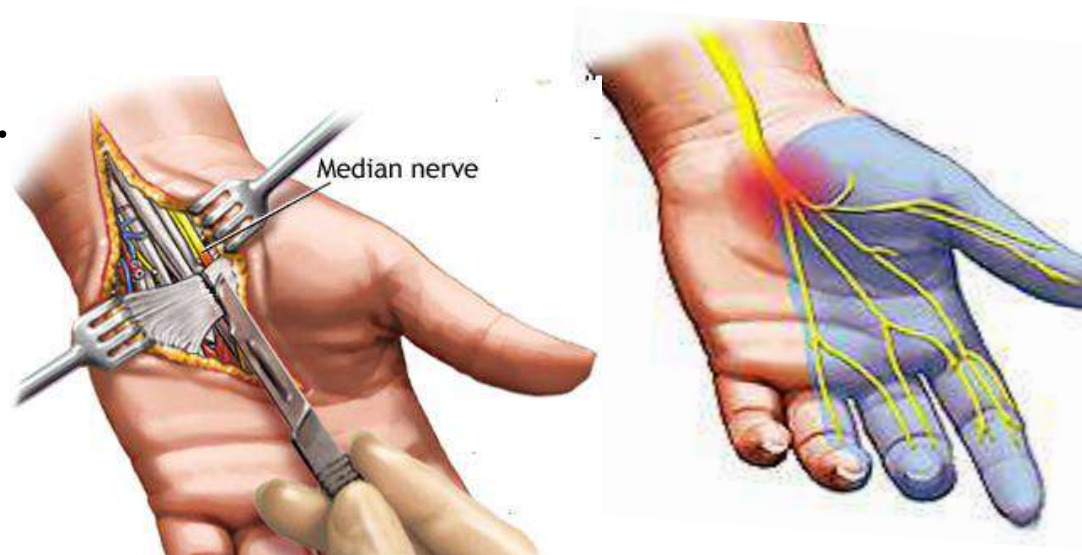
Clinical findings:

Sensory manifestations: Burning, or pins and needles sensation over the anterior surface of the lateral 3.5 finger (**including their nail beds**).

Motor manifestations: Weakness of the thenar muscles.

Treatment:

Decompression of the median n. making a longitudinal incision through the flexor retinaculum.



Ulnar Nerve

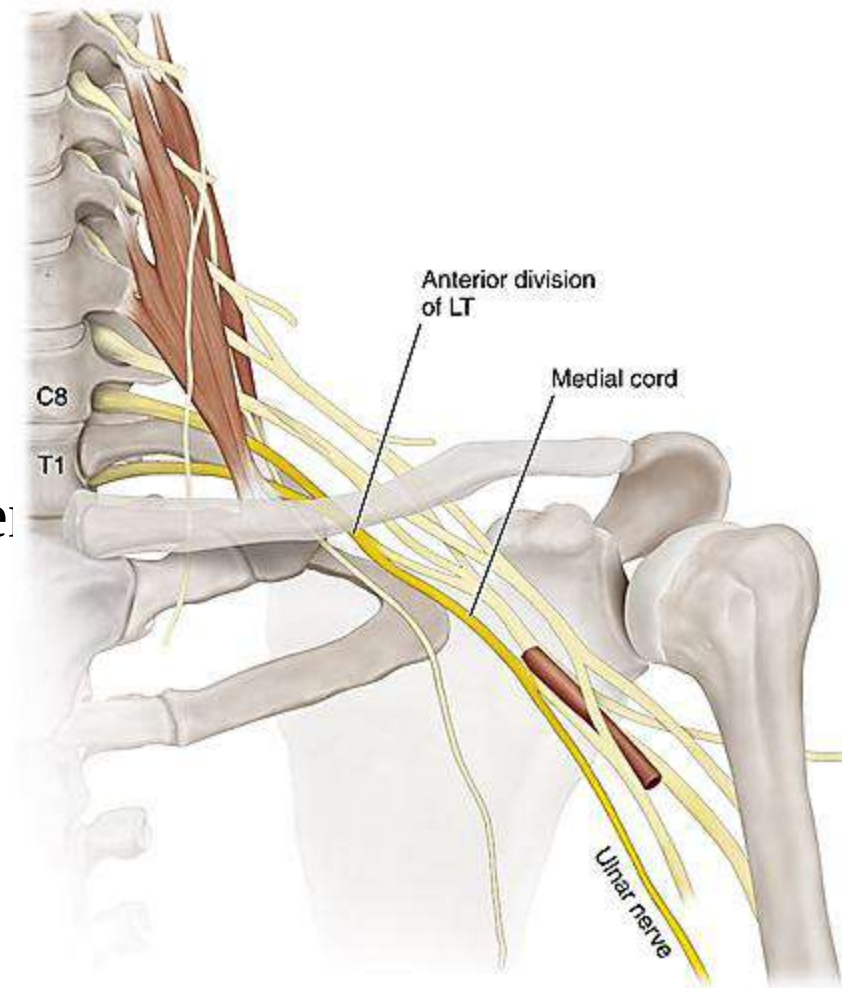
C7,8,T1

Origin:

- From the medial cord of the brachial plexus in the axilla.
- It takes C7 fibers from the median nerve (through a communication in between)

Course and branches :

- In the upper arm
- In the lower arm
- In the forearm
- In the hand



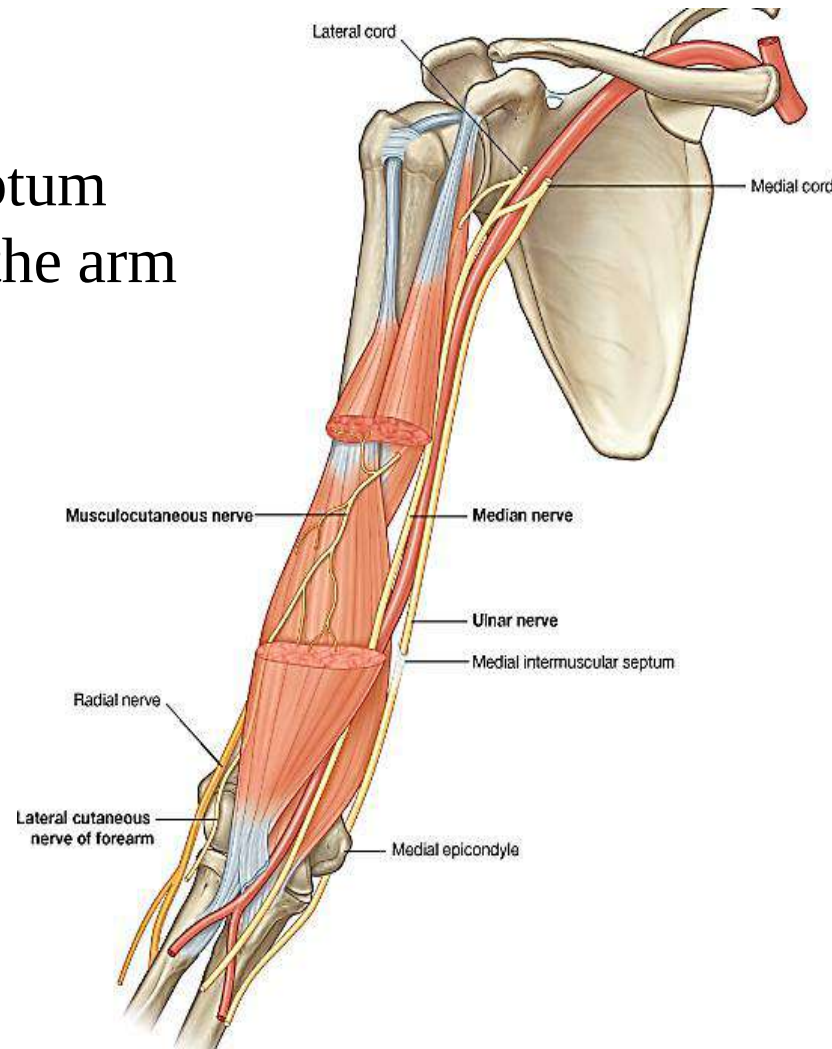
In the upper arm

Course:

- It lies on the medial side of the 3rd part of axillary artery, then upper part of brachial artery.
- It pierces the medial intermuscular septum and enters the posterior compartment of the arm at the middle of the arm.

Branches:

No branches in the axilla and arm.



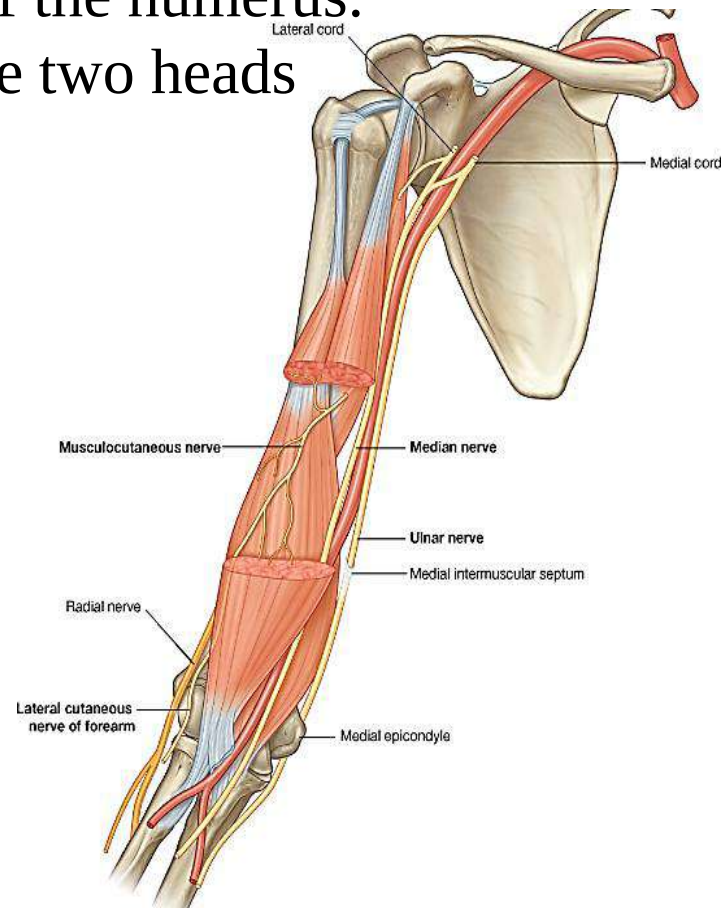
In the Lower arm

Course:

- It enters the posterior compartment of the arm by piercing the medial intermuscular septum at the middle of the arm (with the superior ulnar collateral artery).
- It descends behind the medial epicondyle of the humerus.
- It enters the forearm by passing between the two heads of the flexor carpi ulnaris muscle.

Branches:

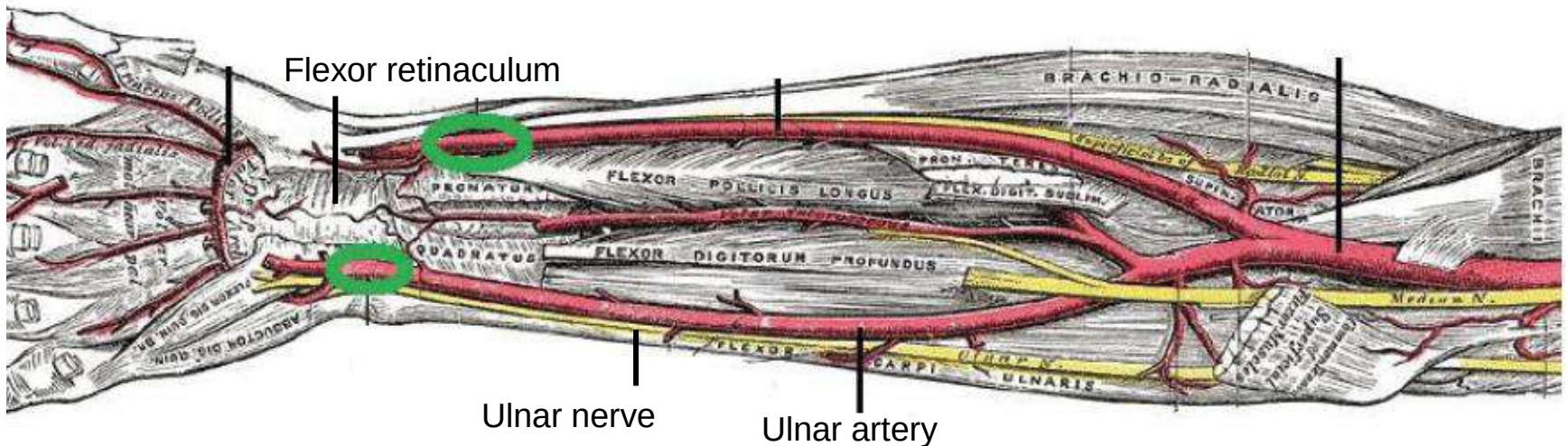
Articular branch to the elbow joint.



In the Forearm

Course:

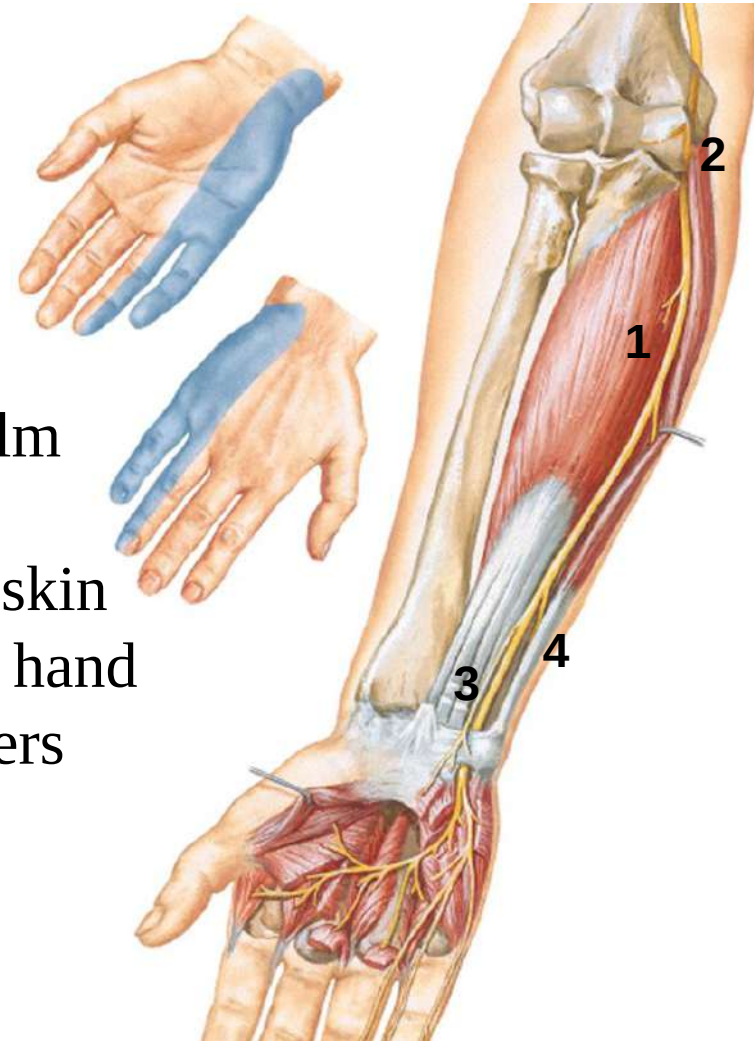
- It crosses the medial collateral ligament of the elbow joint.
- It then descends between the flexor carpi ulnaris and flexor digitorum prof..
- It runs on the medial side of the ulnar artery.
- At the wrist, it becomes superficial, between the tendons of flexor carpi ulnaris and flexor digitorum superf..
- It enters the hand by passing superficial to the flexor retinaculum, between the pisiform (medially) and ulnar artery (laterally).



Branches

In the forearm

1. **Muscular branches:** to flexor carpi ulnaris and medial half of flexor digitorum profundus muscles.
2. **Articular branches:** to the elbow joint.
3. **Palmar cutaneous branches:** passes superficial to the flexor retinaculum and supplies skin of the medial 1/3 of the palm
4. **Dorsal cutaneous branches:** it supplies the skin of the medial 1/3 of the dorsal surface of the hand and the dorsal surface of the medial 1.5 fingers (except nail beds).



In the hand

Course: It enters the hand superficial to the flexor retinaculum.

Branches

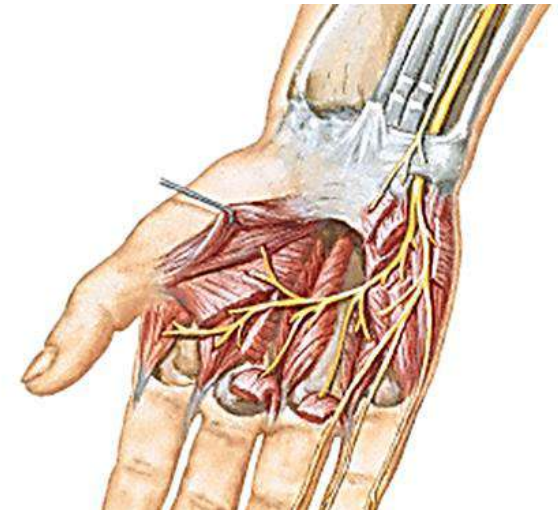
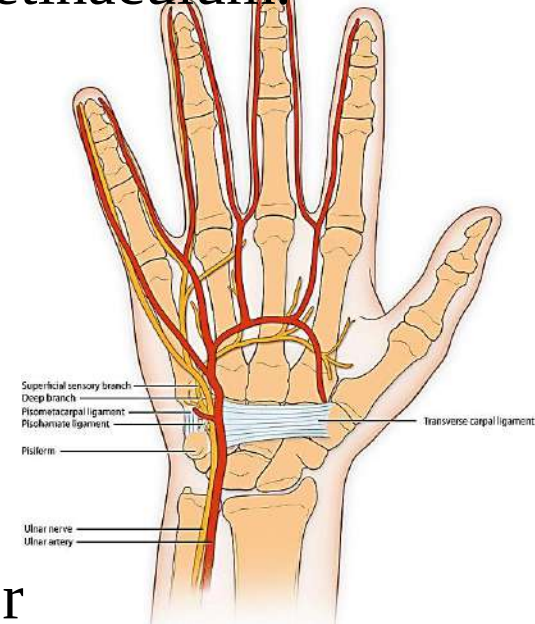
Superficial branch:

- supplies palmaris brevis muscles
- give cutaneous innervation to palmar surface of the medial 1.5 fingers and their nail beds.

Deep branch: then turns laterally between the flexor tendons and the interossei, in the concavity of the deep palmar arch.

It gives muscular branches to

- the 3 hypothenar muscles,
- all interossei,
- adductor pollicis
- the medial 2 lumbricals.

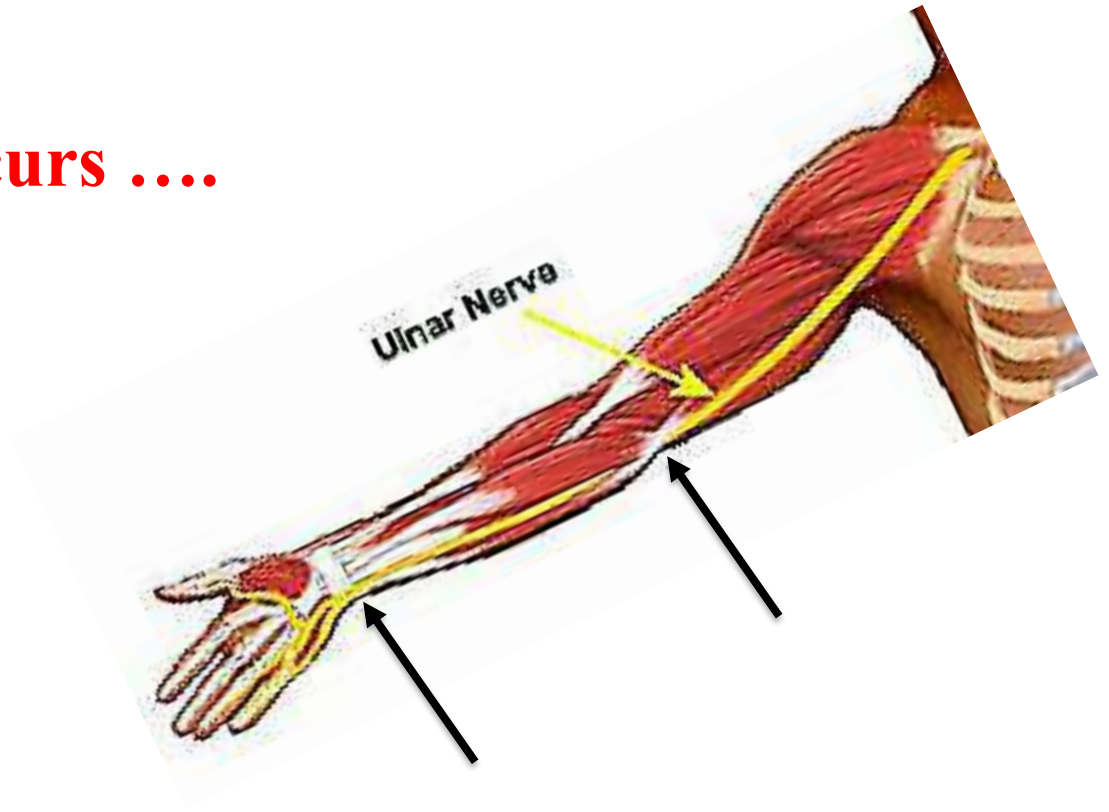


Ulnar Nerve

Lesion usually occurs

1- at the elbow

2-at the wrist



1. lesion at the elbow

Causes: fracture **medial epicondyle of humerus**.

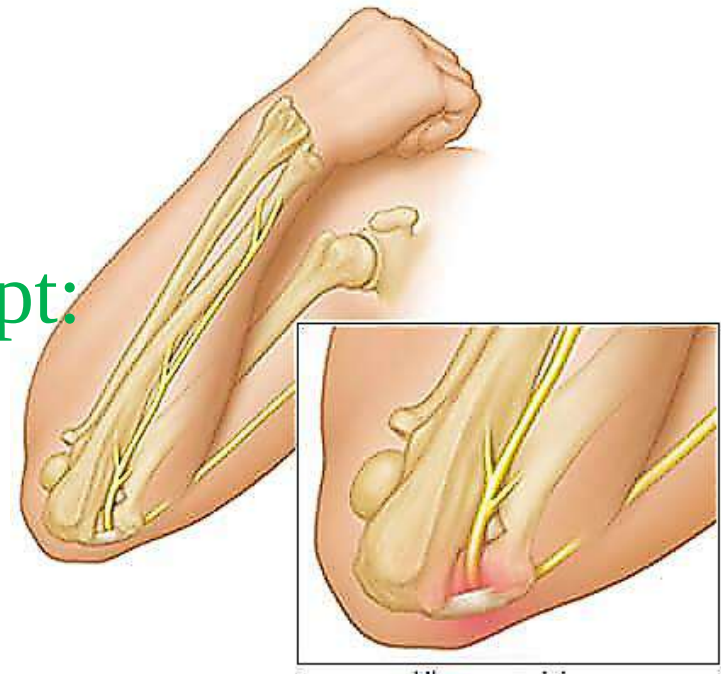
Muscles affected: flexor carpi ulnaris, medial half of flexor digitorum profundus, palmaris brevis, hypothenar muscles, all interossei, medial 2 lumbricals and adductor pollicis.

Motor affection:

No abduction or adduction of fingers,

No adduction the thumb.

Wasting of small muscles of the hand **except:**
the thenar muscles & 1,2 lumbricals



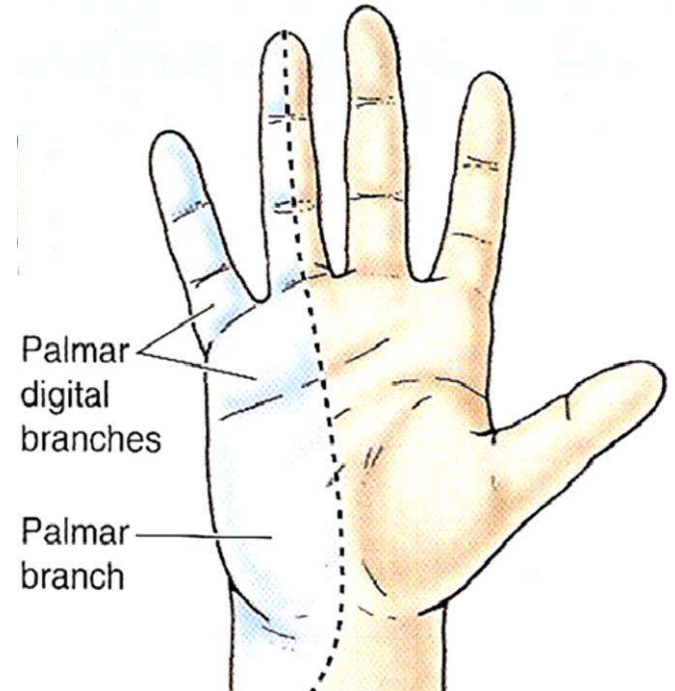
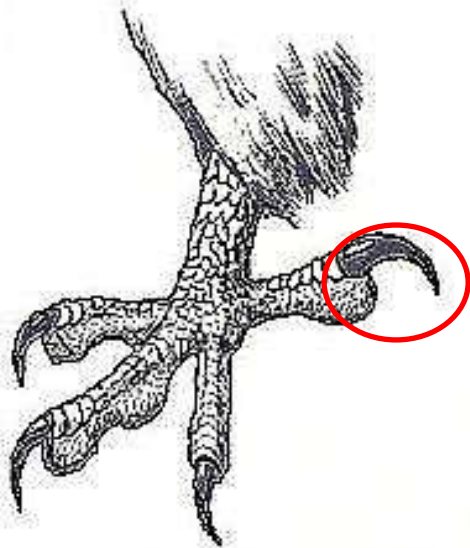
Deformity: *claw hand” deformity.*

Extension of metacarpophalangeal joints of fingers 4,5

Flexion of interphalangeal joints more on the 4, 5 fingers

Sensory loss: of the medial 1/3 of the palmar and dorsal surfaces of the hand and medial 1.5 fingers.

Vasomotor changes: warming and drying of the medial 1/3 of the palmar and dorsal surfaces of the hand and medial 1.5 fingers.



2. Lesion at the wrist

Causes: stab or cut wound at the wrist.

Muscles affected:

palmaris brevis, hypothenar muscles, all interossei, medial 2 lumbricals
adductor pollicis.

Motor affection:

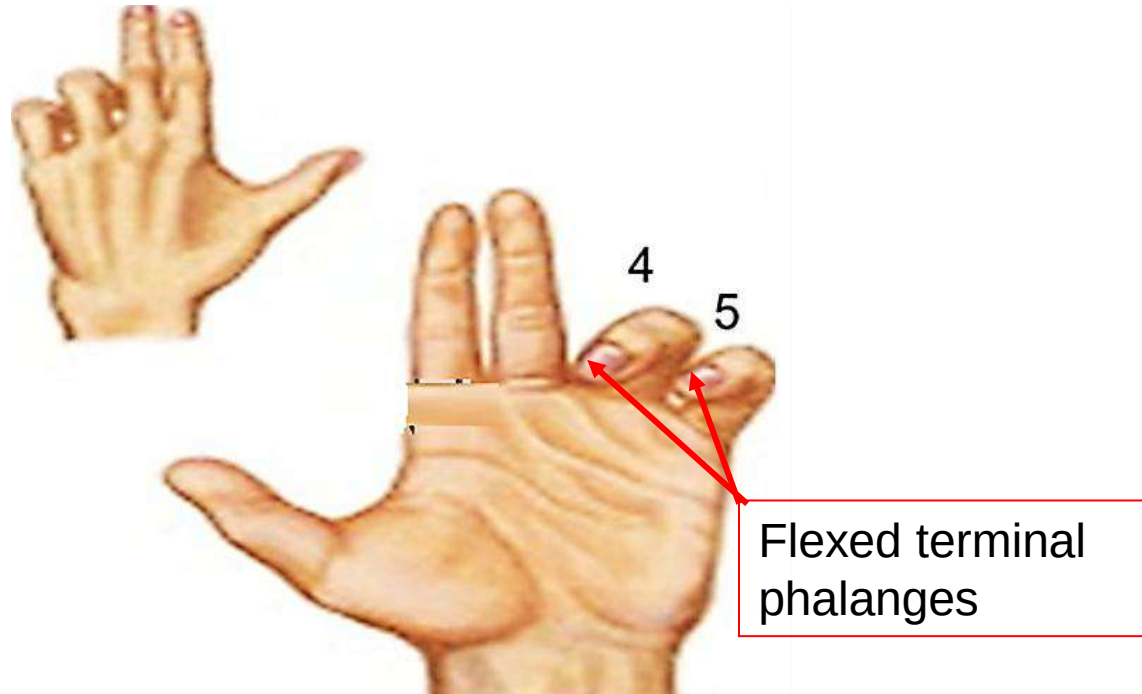
- the patients can not abduct and adduct his fingers, or adduct the thumb.
- Wasting of small muscles of the hand except the thenar muscles & lumbricals 1,2.

Deformity: is a **“severe claw hand” deformity.**

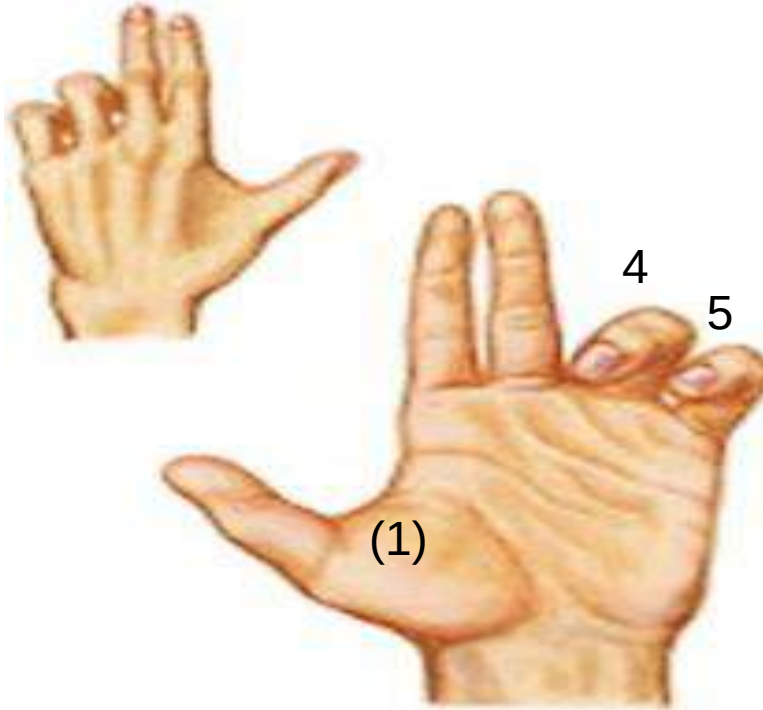
Severe claw hand because the flexor digitorum profundus is spared resulting in flexion of distal phalanges.

Sensory loss: of the medial 1/3 of the **palm** and anterior surface of the medial 1.5 fingers and their nail beds.

Vasomotor changes: warming and drying of the medial 1/3 of the palm and anterior surface of the medial 1.5 fingers.



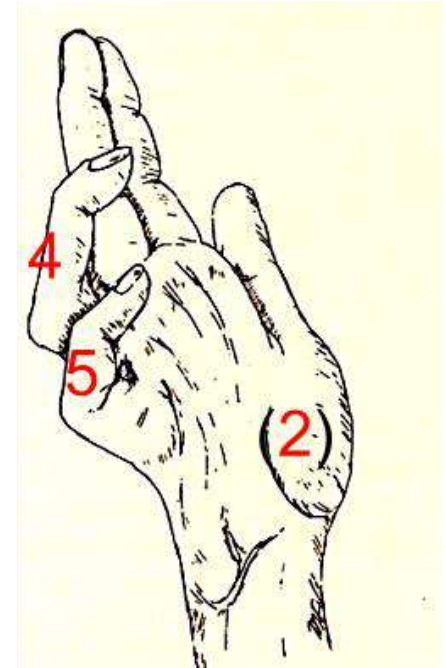
(1) Severe Claw hand



Lesion at the wrist

- medial $\frac{1}{2}$ of flexor digitorum profundus is spared
- distal phalanx is flexed

(2) Partial Claw hand



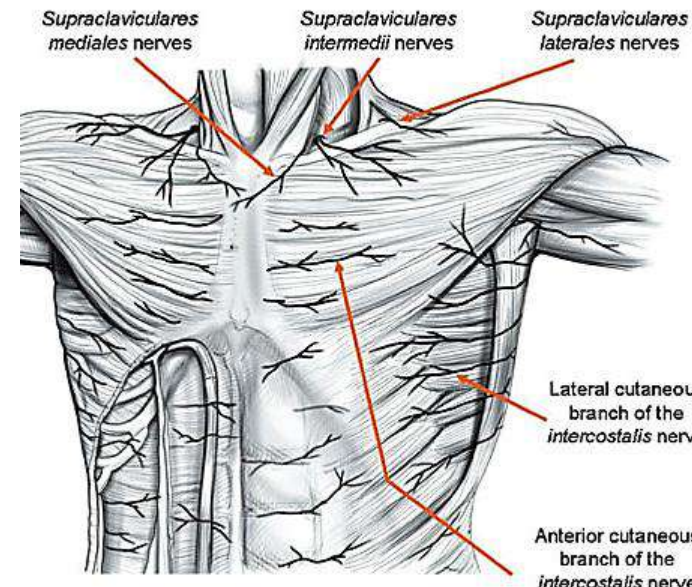
Lesion at the elbow

- medial $\frac{1}{2}$ of flexor digitorum profundus affected
- distal phalanx is extended

Cutaneous innervation of the Upper limb

Pectoral Region, Axilla and Back

- **Supraclavicular nerves** (C3,4) from cervical plexus: skin of the anterior aspect of the pectoral region till the sternal angle (2nd costal cartilage).
- **Anterior cutaneous branches of the intercostal nerves** skin from the midline to the mid-clavicular line.
- **Lateral cutaneous branches of the intercostal nerves** skin from the lateral side of the pectoral region.
- **Intercostobrachial nerve** (T2): skin of the axilla.
- **Dorsal rami of the spinal nerves (T1-12):** skin of the back.



The Arm

1. Supraclavicular nerves (C3,4) from cervical plexus:

- skin over the shoulder and upper half of deltoid.

2. Upper lateral cutaneous nerve of the arm (C5,6) from the axillary nerve:

- skin over the lower half of the deltoid.

3. Lower lateral cutaneous nerve of the arm (C5-T1): from the radial nerve

- skin of the lateral surface of the arm below the deltoid.

4. Medial Cutaneous nerve of the arm (C8,T1):

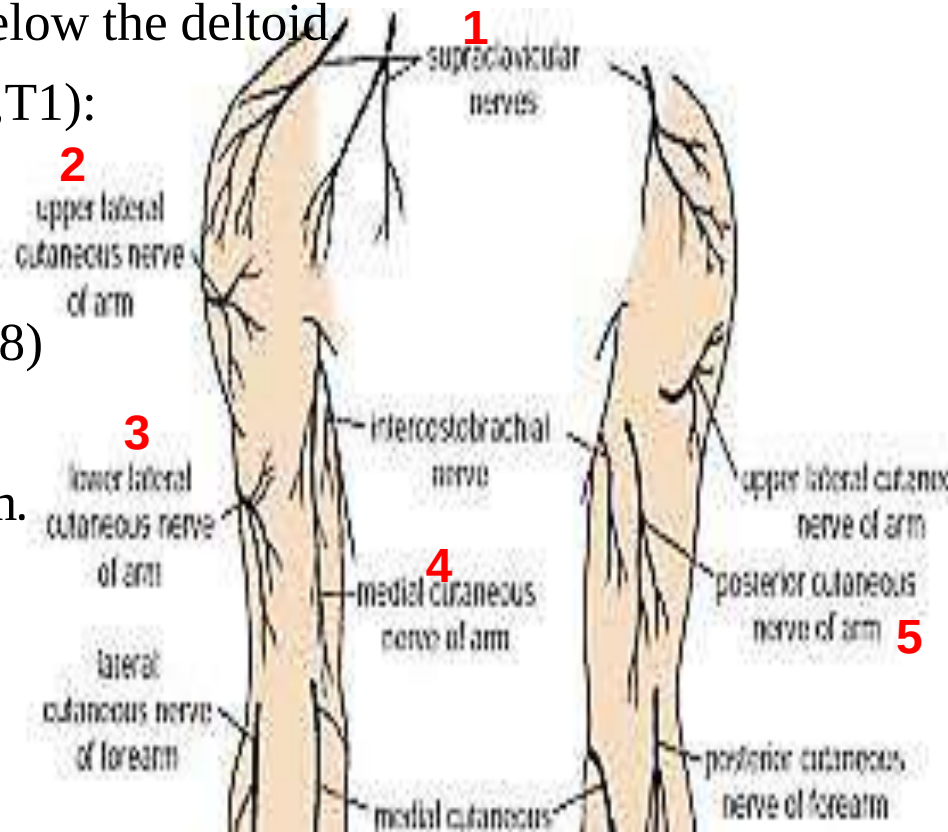
from the medial cord

- skin of the medial side of the arm.

5. Posterior cutaneous nerve of the arm (C8)

from the radial nerve .

- skin of the posterior surface of the arm.



The Forearm

1. Lateral cutaneous nerve of the forearm

the continuation of the musculocutaneous nerve,

- the lateral side of the forearm.

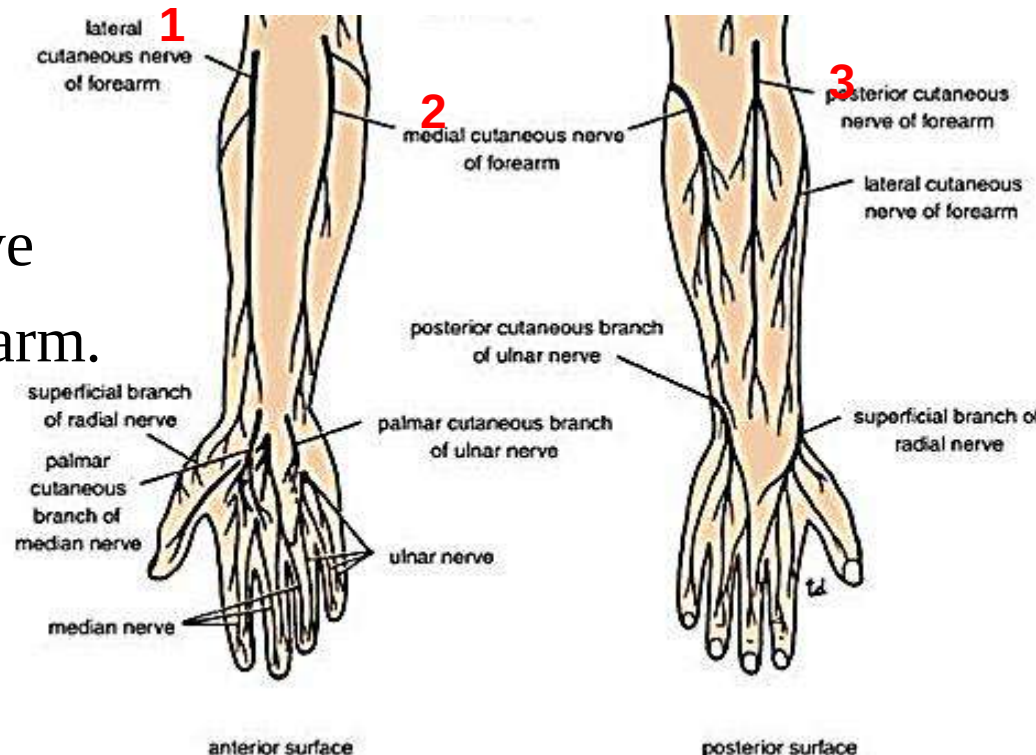
2. Medial Cutaneous nerve of the forearm

from the medial cord of brachial plexus

- the medial side of the forearm.

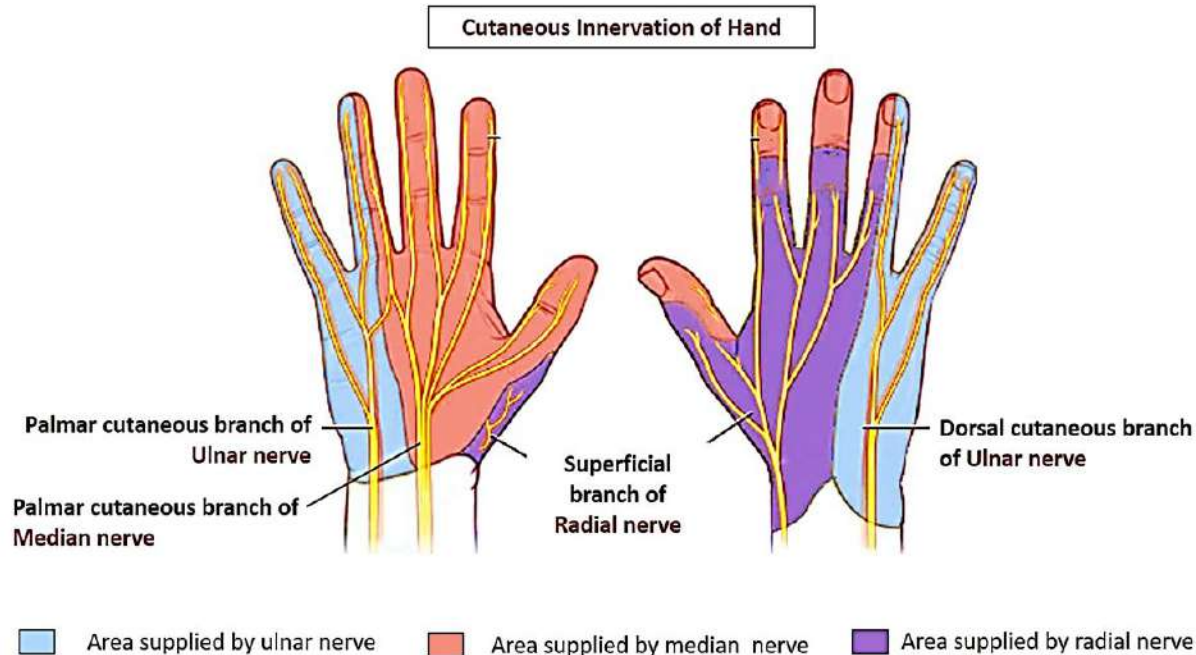
3. Posterior cutaneous nerve of the forearm: from the radial nerve

- the posterior aspect of the forearm.



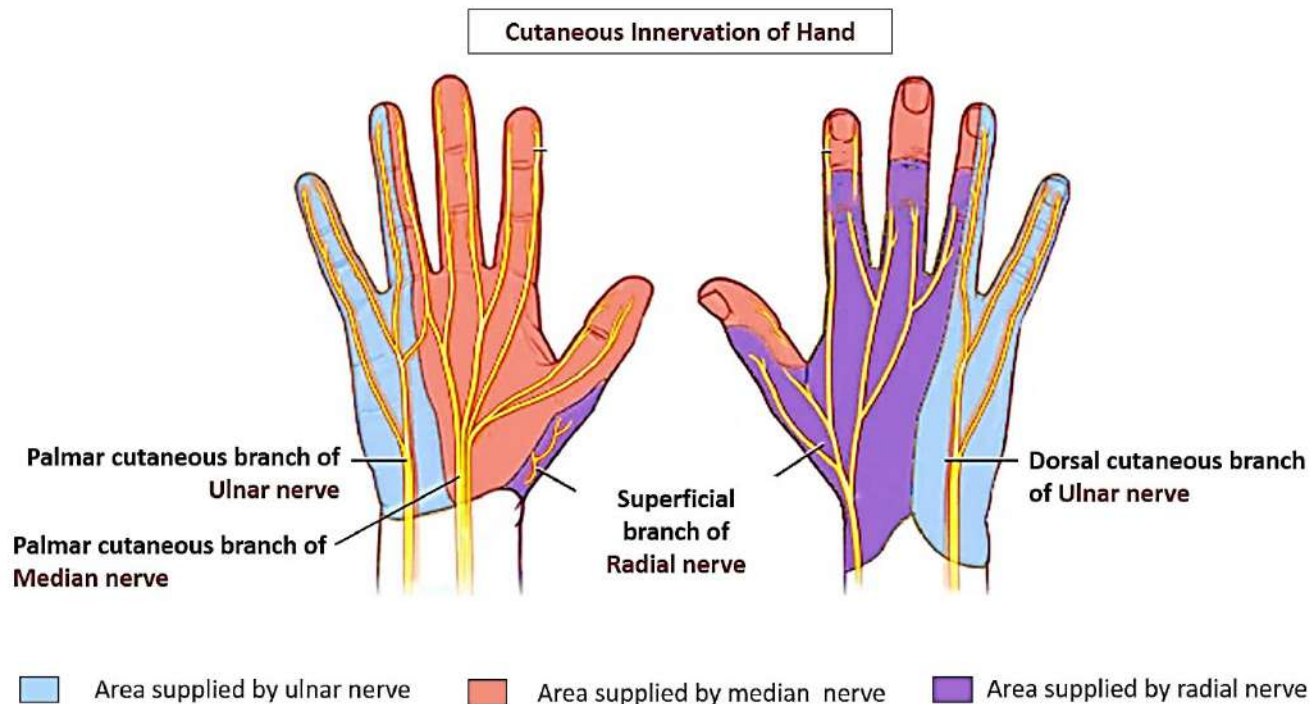
Palm of hand

- **lateral 2/3 of the palm:** by the palmar cutaneous branch of median nerve.
- **medial 1/3 of the palm:** by the palmar cutaneous branch of ulnar nerve.
- **lateral 3.5 fingers:** by the cutaneous branches of the median nerve.
- **medial 1.5 finger:** by the cutaneous branches of the ulnar nerve.

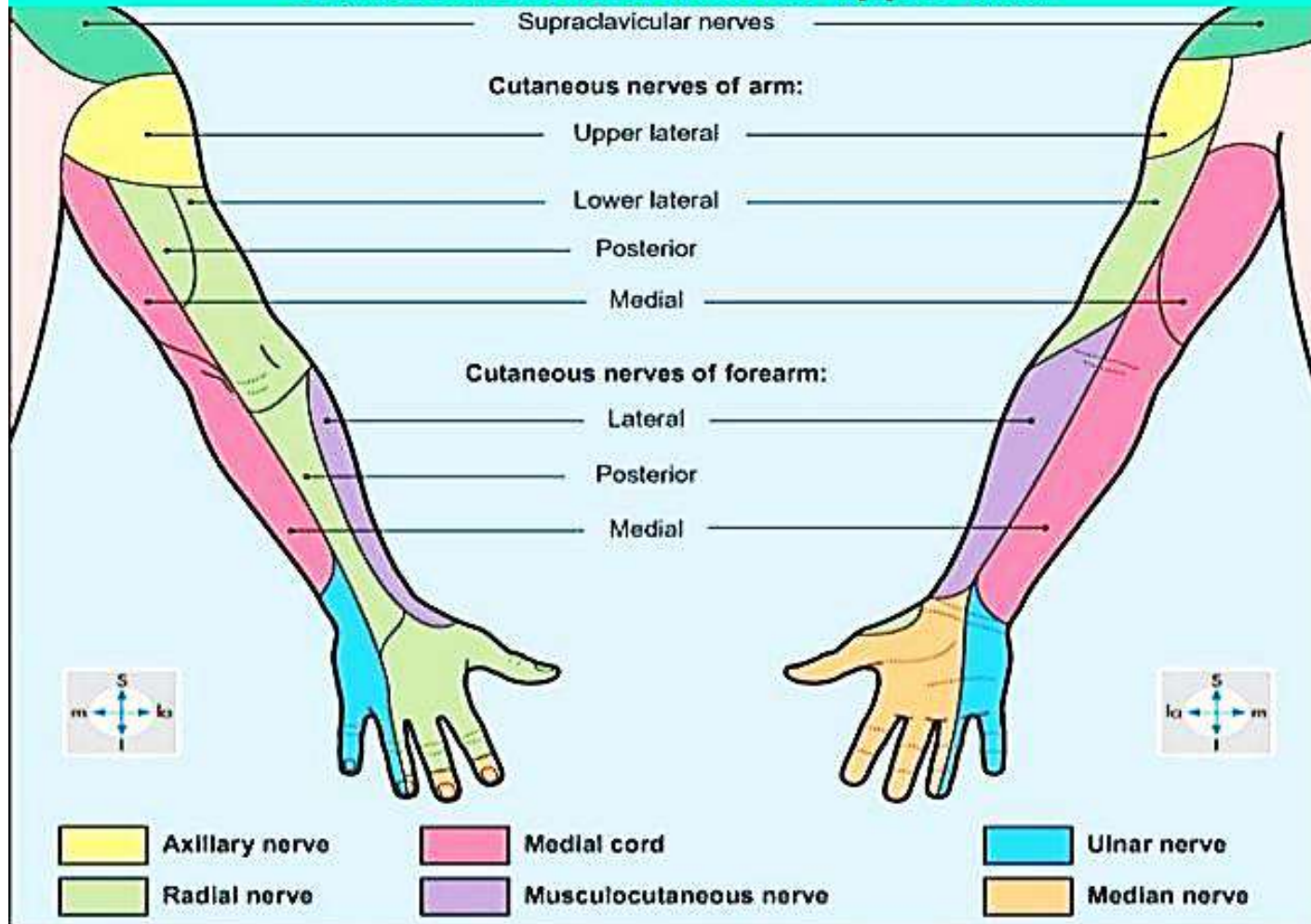


Dorsum of hand

1. **Superficial branch of radial nerve:** for the lateral 2/3 of the dorsum of the hand and the lateral 3.5 fingers (except nail beds).
2. **Dorsal cutaneous branch of the ulnar nerve:** for the medial 1/3 of the dorsum of the hand and the medial 1.5 fingers (except nail beds).

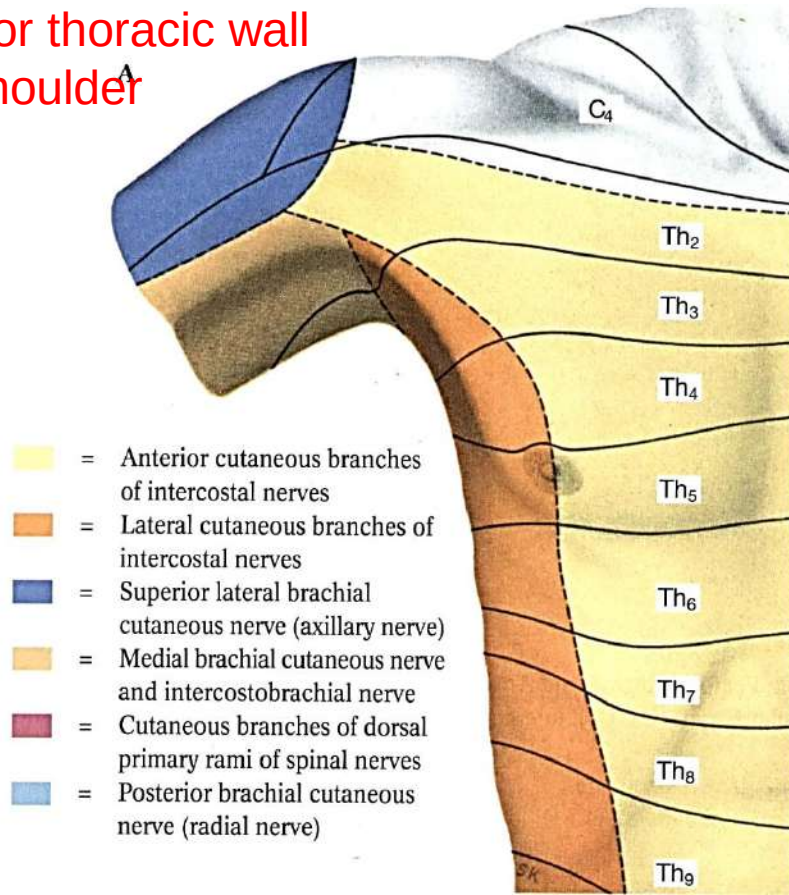


Cutaneous innervation of upper limb

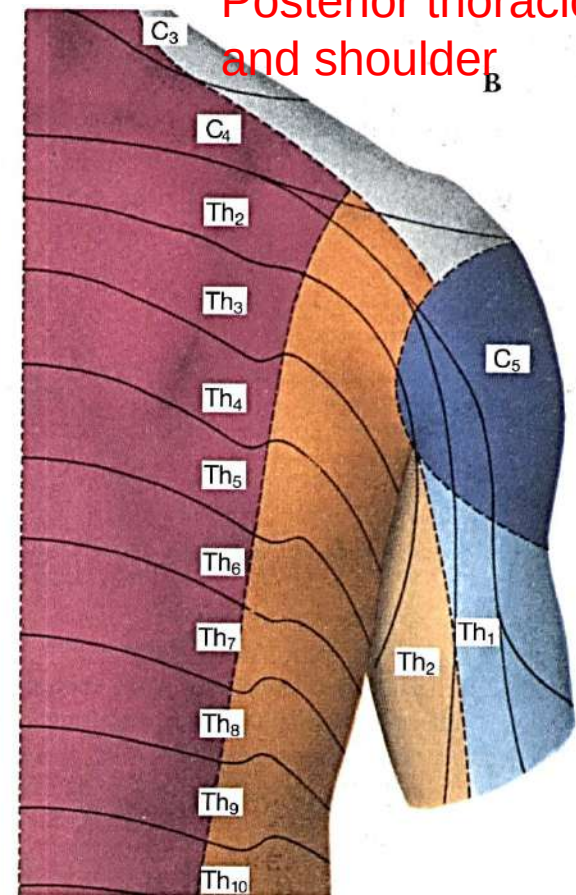


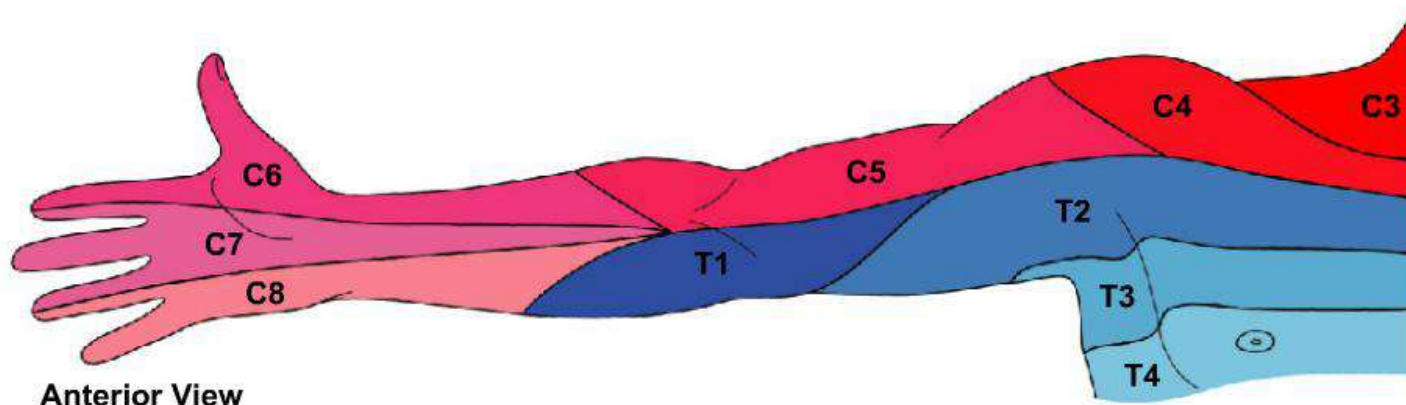
Dermatomes

Anterior thoracic wall
and shoulder

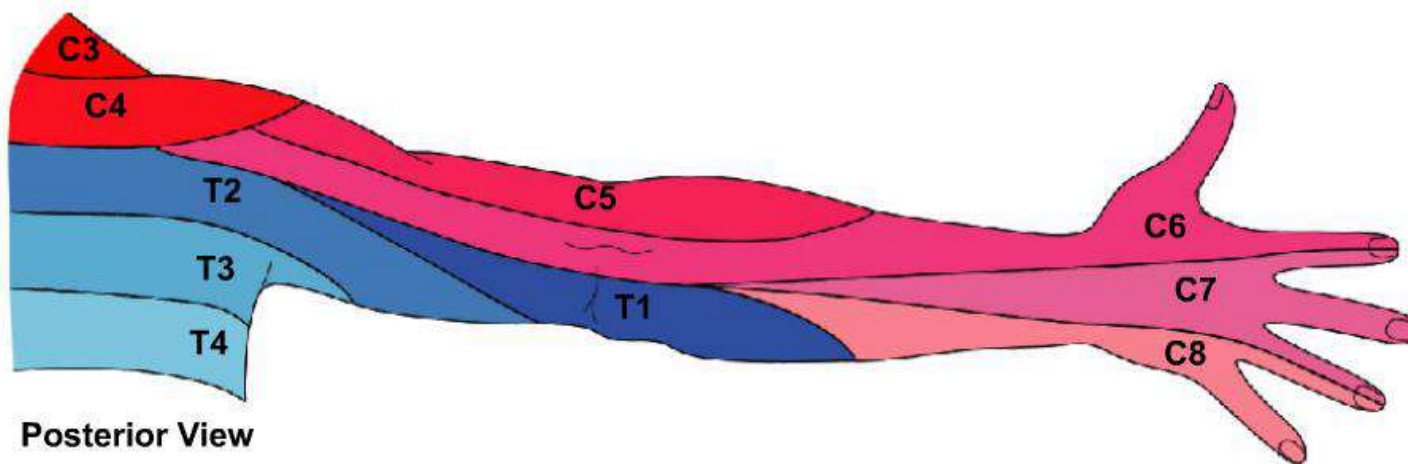


Posterior thoracic wall
and shoulder





Anterior View



Posterior View

Remember !

- Lesion of radial nerve... wrist drop
- Lesion of posterior interosseous nerve.....no wrist drop
- Lesion of ulnar nerve at the elbow... claw hand
at the wrist...severe claw hand
- Lesion of median nerve.... Ape hand
- Klumpk palsy (lower trunk).....claw hand
- Erb Duchenne Palsy (upper trunk).... Porter tip position

GOOD LUCK

Thank you